Independent Research Project

Evaluation of particle dispersion characteristics over flat field and idealized building array using Lattice Boltzmann method

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Nomenclature

Footprint and Source Area Theory

η Measured scalar concentration or flux

Ω_P Source area of level P

Φ_{tot} Total influence (integral of the source weight function)

f Source weight function (footprint function)

L Obukhov length (atmospheric stability parameter) [m]

P Fractional contribution to total measured signal

Q Source strength or emission rate

Physical and Flow Parameters

ν Kinematic viscosity of air [m^2/s]

H Reference building height [m]

k Turbulent Kinetic Energy (TKE) [$\mathrm{m}^2/\mathrm{s}^2$]

Re Reynolds number

u, v, w Velocity components in x, y, z directions [m/s]

U_H Mean wind velocity at building height [m/s]

$u_{rms}, v_{rms}, w_{rms}$ Root-mean-square velocity fluctuations [m/s]

z_0 Aerodynamic roughness length [m]

Statistical and Validation Metrics

D Relative error tolerance (typically 0.25)

$FAC2$ Factor of two (fraction of data within a factor of two)

MAE Mean Absolute Error

O_i Observed value from AIJ experimental data

P_i Predicted value from LBM simulation

q Hit rate (fraction of predictions within tolerance)

$RMSE$ Root Mean Square Error

W Absolute error tolerance (typically 0.05)

1 Introduction

1.1 Background

Air pollution in urban areas represents an ongoing global challenge that negatively impacts individual health, ecosystem integrity, and places increasing pressure on medical systems and economies worldwide. Urban air pollution comprises multiple components, including carbon dioxide (CO_2), carbon monoxide (CO), and particulate matter with an aerodynamic diameter of less than 2.5 μm ($PM_{2.5}$).

The health implications are severe; recent data from the World Health Organization (2024) indicates that ambient air pollution contributes to approximately 4.2 million premature deaths globally each year. While mortality is largely attributed to increased private vehicle demand and population growth, a significant contributing factor is the lack of consideration for pollutant dispersion during rapid urban development (Li et al., 2021). As noted by Li et al. (2021), ”hasty and irrational construction strategies” often lead to poor ventilation, trapping pollutants in high concentrations near the ground level.

Understanding and mitigating these concentrations requires a dual-level approach. First, it involves the utilization of mechanical factors, which include forced convection by wind, natural convection by solar radiation, and traffic-induced turbulence. Second, it requires the strategic design of urban morphology. This focus on the ”sound design of the urban landscape” emphasizes building geometries and urban density to improve the city’s natural capacity to flush out pollutants (Li et al., 2021).

Because pollutant dispersion is governed by complex wind patterns, accurate prediction of how these pollutants behave in the built environment has become a critical research topic. While traditional research has relied on physical measurements and empirical models, Computational Fluid Dynamics (CFD) has emerged as a superior alternative for these problems. As highlighted by Shirzadi and Tominaga (2024), CFD is advantageous as it provides comprehensive data for the entire flow and dispersion field at a full physical scale, and is not restricted by the complex ”similarity requirements” that physical wind tunnel replicas must follow to ensure accuracy.

Additionally, when assessing urban air quality, it is essential to distinguish between ”near-field” and ”far-field” dispersion phenomena. As defined by Tominaga and Stathopoulos (2024), near-field dispersion occurs in the immediate vicinity of an emitting building—typically within a few hundred meters. In this region, pollutant behavior is dominated by the complex interaction between the atmospheric flow and the building’s specific geometry, often leading to localized high concentrations in surrounding streets or on building surfaces.

In contrast, far-field phenomena are characterized by larger horizontal length scales where the influence of individual buildings becomes relatively minor compared to general atmospheric motion. Historically, near-field problems were addressed using ”fast response” operational models, such as the Gaussian plume model, or semi-empirical design tools like the ASHRAE model (Tominaga & Stathopoulos, 2024). However, these models often rely on wake-averaged flow values and cannot explicitly treat the detailed plume behavior as it interacts with building obstacles.

This limitation necessitates the use of micro-scale Computational Fluid Dynamics (CFD). By solving the transport equations for concentration alongside the Navier-Stokes equations for velocity, CFD provides the high-resolution data required to identify ”street-scale” pollution hotspots. While more computationally intensive than operational models, the high-fidelity results provided by the LBM-LES framework on TSUBAME 4.0 allow for the detailed sensitivity and uncertainty analysis recommended by current best-practice guidelines (Tominaga and Stathopoulos, 2024, Yokouchi et al., 2020).

1.2 Objective

2 Theory

2.1 Lattice Boltzmann Method

To simulate urban air pollutant transport, this study employs the Lattice Boltzmann Method (LBM), a mesoscopic computational fluid dynamics (CFD) technique. Unlike traditional solvers that directly discretize the macroscopic Navier-Stokes equations, LBM models fluid flow by simulating the collective behavior of a distribution of particles on a discrete grid, or ”lattice” (Krüger et al., 2017).The movement of these particles is governed by the Lattice Boltzmann Equation (LBE), which describes the evolution of the discrete-velocity distribution function f_i:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = \Omega_i(f) + S_i \tag{1}$$

Where f_i is the distribution function representing the density of particles moving with velocity components $\mathbf{c}_i$ in the i-th direction. The term Ω_i represents the collision operator, which models the internal redistribution of particle momentum, while S_i is the source term. The inclusion of S_i is critical for this research as it allows for the integration of external body forces and the modeling of mass sources, such as vehicle emissions.The simulation proceeds through a sequence of two primary operations: streaming and collision. During the streaming step, particles move from their current node $\mathbf{x}$ to adjacent points based on their respective velocities. Following this, the collision step redistributes the particle velocities at each node based on local interactions, effectively mimicking fluid viscosity and relaxation toward equilibrium. As noted by Krüger et al. (2017), LBM offers several distinct advantages for simulating complex urban environments. First, the underlying Boltzmann equation is essentially a hyperbolic advection equation where the source terms depend only on local values of f. Because these operations rely strictly on information from a node and its immediate neighbors, the algorithm is inherently local. This locality facilitates high-performance parallelization, which is essential for large-scale urban simulations. Furthermore, the grid-based nature of LBM allows for the efficient handling of complex building geometries and irregular urban morphologies without the need for the intensive re-meshing often required by finite-element methods.

2.1.1 Lattice Configuration and Accuracy

The choice of the D3Q27 lattice model is fundamental to the accuracy of the turbulence modeling in this research (Yokouchi et al., 2020). While the simpler D3Q19 model is often used to reduce computational costs, it lacks the rotational invariance required to capture complex secondary flows accurately. As demonstrated by Suga et al. (2015), the D3Q27 lattice provides a superior representation of turbulent structures. Their comparative analysis of turbulent pipe flow showed that the D3Q27 model produces significantly more realistic flow fields than the D3Q19 model, which tends to struggle with resolving fine-scale turbulent fluctuations in three-dimensional space. By utilizing 27 discrete velocities, the model can more effectively resolve the secondary circulations and high-frequency fluctuations necessary for an in-depth understanding of the pollutant dispersion mechanism in urban areas.The internal redistribution of particle momentum is governed by the Bhatnagar-Gross-Krook (BGK) collision operator. This operator assumes that the particle populations relax toward a local equilibrium state, f_i^{eq}, at a rate controlled by a single relaxation time τ (Krüger et al., 2017):

$$\Omega_i(f) = -\frac{f_i - f_i^{eq}}{\tau} \Delta t \tag{2}$$

The equilibrium distribution f_i^{eq} is formulated as a second-order expansion of the Maxwell-Boltzmann distribution in terms of the macroscopic density ρ and velocity $\mathbf{u}$:

$$f_i^{eq}(\mathbf{x}, t) = w_i \rho \left(1 + \frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} + \frac{(\mathbf{u} \cdot \mathbf{c}_i)^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right) \tag{3}$$

Here, c_s represents the isothermal speed of sound, defined as $c_s^2 = \frac{1}{3} \frac{\Delta x^2}{\Delta t^2}$. This formulation ensures the local conservation of mass and momentum while establishing the relationship between pressure and density as $p = c_s^2 \rho$.

2.1.2 Large Eddy Simulation and Turbulence Modeling

To capture the transient, multi-scale nature of urban wind flows, this research utilizes a Large Eddy Simulation (LES) approach. The unresolved subgrid-scale (SGS) motions are accounted for using the Coherent-structure Smagorinsky Model (CSM). In a standard Smagorinsky framework, the turbulent viscosity ν_t is determined by the strain-rate tensor $|S|$ and a filter width Δ:

$$\nu_t = (C_S \Delta)^2 |S| \tag{4}$$

Within the LBM framework, the total viscosity ν (the sum of molecular and turbulent viscosity) is directly linked to the relaxation time:

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2} \right) \tag{5}$$

While a standard Smagorinsky constant C_S typically ranges between 0.1 and 0.2 (Blazek, 2015), the CSM used here employs a dynamic parameter based on the second invariant of the velocity gradient tensor. This approach allows the model to automatically account for wall-damping effects, ensuring more accurate flow predictions near building surfaces without requiring manual damping functions (Phuc et al., 2018).

2.1.3 Implementation of Lagrangian Stochastic Model

To simulate the transport and dispersion of particulate matter within the urban environment, this study utilizes a Lagrangian Stochastic Model (LSM). Unlike Eulerian methods that treat pollutants as a continuous concentration field, the LSM tracks the trajectories of thousands of discrete ”passive” particles within the turbulent flow field generated by the LBM. This approach is particularly effective for modeling dispersion in the inhomogeneous and non-Gaussian turbulence characteristic of the urban planetary boundary layer (PBL). By simulating a statistically significant ensemble of particle paths, the ensemble-mean concentration can be derived, which is directly proportional to the particle number density or the probability density function (PDF) of particle positions (Weil et al., 2004).The trajectory of each particle is determined by integrating its position x_p over time. For a particle starting at an initial position x_0, the position at the next time step $t + \Delta t$ is calculated as:

$$x_p(x_0, t + \Delta t) = x_p(x_0, t) + u_p(x_0, t)\Delta t \tag{6}$$

where u_p represents the instantaneous Lagrangian particle velocity. This velocity is composed of two primary components:

$$u_p = u_r + u_s$$

In this formulation, u_r is the resolved grid-scale (GS) velocity obtained directly from the LBM-LES results, representing the large-scale atmospheric motions. The term u_s represents the subgrid-scale (SGS) velocity fluctuations, which account for the smaller, unresolved turbulent motions.The resolved velocity u_r at the particle's location is determined by interpolating the LBM grid results using the weighting factors w_i of the neighboring nodes:

$$u_r = \sum_{i=0}^{7} (w_i U_i)$$

To account for the unresolved fluctuations, the SGS velocity u_s is modeled as a stochastic process. Following the framework established by Weil et al. (2004) and Thomson (1987), the evolution of the SGS velocity du_s is described by a stochastic differential equation:

$$du_s = -\frac{3 f_s C_0 \epsilon}{4} dt + \frac{1}{2} \left(\frac{1}{e_s} \frac{de_s}{dt} u_s + \frac{2}{3} \nabla e_s \right) dt + (f_s C_0 \epsilon)^{1/2} d\xi \tag{7}$$

where e_s is the local SGS turbulent kinetic energy (TKE) and f_s represents the mean contribution of the SGS TKE to the total TKE. The term ϵ denotes the total dissipation rate, while C_0 is a universal constant (set to 4.0 in this study). The stochastic component $(f_s C_0 \epsilon)^{1/2} d\xi$ incorporates Gaussian white noise via the displacement vector $d\xi$, allowing the model to represent the random nature of sub-lattice turbulent dispersion.

2.1.4 Advanced Wall Boundary Conditions

In LBM, the standard Bounce-Back (BB) scheme assumes that fluid particles hitting a solid boundary reverse their momentum, effectively enforcing a no-slip condition ($\mathbf{u} = 0$) at the wall. While numerically stable and simple to implement, the BB scheme requires the first grid point to reside within the viscous sublayer to accurately capture wall shear stress. In urban-scale simulations, the grid resolution is typically much larger than the thickness of this sublayer, causing the BB scheme to underpredict shear drag.To resolve this, a Wall Function Boundary (WFB) is implemented, following the approach evaluated by Han et al. (2021). The WFB utilizes Spalding's Law to model the non-linear velocity profile in the logarithmic layer, allowing the simulation to approximate near-wall velocity without fully resolving the thin viscous region. This modification is applied during the collision step by introducing a momentum loss term based on the wall shear stress τ_w:

$$f_a^* = f_a \pm \frac{\Delta t}{2\Delta y} \tau_{w,j} \tag{8}$$

where $\tau_{w,j}$ is the shear stress in the j-direction (x or z). The wall shear stress is determined by solving Spalding's implicit equation:

$$y^+ = u^+ + e^{-\kappa B} \left[e^{\kappa u^+} - 1 - (\kappa u^+) - \frac{(\kappa u^+)^2}{2} - \frac{(\kappa u^+)^3}{6} \right] \tag{9}$$

Here, y^+ is the dimensionless distance from the wall, and u^+ is the dimensionless velocity. By calculating the friction velocity $u_\tau = \sqrt{\tau_w / \rho}$ from this relationship, the model applies a "reverse resistance" that decelerates particles near building surfaces. As demonstrated by Han et al. (2021), this WFB approach significantly improves the accuracy of shear drag predictions in high-Reynolds-number urban flows where coarse grids are a necessity.

2.1.5 Subgrid-Scale Turbulent Kinetic Energy (SGS TKE)

The calculation of the SGS TKE (e_s) is vital for the Lagrangian Stochastic Model, as it dictates the magnitude of the stochastic velocity fluctuations. While traditional LES often parameterizes e_s via the eddy viscosity and a mixing length ($v_t = c_k l e_s^{1/2}$), this study utilizes a filtering approach as implemented by Suga et al. (2015) and Yokouchi et al. (2020). The SGS TKE (k_{sgs}) is derived from the difference between the instantaneous grid-scale velocity and a filtered velocity field:

$$k_{sgs} = C_{kes} \sum_{i=1}^{3} (\langle \hat{U}_i \rangle - \langle U_i \rangle)^2 \tag{10}$$

The filtered velocity $\langle \hat{U}_i \rangle$ is calculated using a weighted average of the central node (50% weight) and its six immediate neighbors (1/12 weight each):

$$\langle \hat{U}_i \rangle = \frac{1}{2} \langle U_i \rangle + \frac{\langle U_i^E \rangle + \langle U_i^W \rangle + \langle U_i^N \rangle + \langle U_i^S \rangle + \langle U_i^T \rangle + \langle U_i^B \rangle}{12} \tag{11}$$

This local filtering operation is highly compatible with the LBM's neighbor-only communication pattern, maintaining the algorithm's high parallel efficiency while providing a robust estimate of local turbulence intensity.

2.1.6 Stochastic Integration and Stability

The stochastic component of the LSM involves the integration of Gaussian white noise, $\xi(t)$, which is formally defined as the time derivative of a Wiener process $W(t)$:

$$W_i(t) = \int_0^t \xi(s) ds \tag{12}$$

In accordance with Brownian motion theory, the variance of this process increases linearly with time. Given that the SGS velocity u_s is modeled as a random walk, the stochastic term $(f_s C_0 \epsilon)^{1/2} d\xi$ can occasionally produce unrealistically high instantaneous velocities. This is particularly prevalent near building walls where e_s is high and the gradient terms in the LSM equation are steep. To prevent numerical instability and maintain physical consistency, a velocity limitation is enforced, ensuring that the total Lagrangian particle velocity u_p remains within physically plausible bounds relative to the local mean flow Yokouchi et al., 2020.

2.1.7 Model Selection and Validation via Flat Plate Analysis

Before applying the model to complex urban morphologies, Yokouchi et al. (2020) conducted a validation study using particle dispersion over a flat plate to identify the most accurate combination of numerical schemes. The evaluation compared three primary configurations: (1) a baseline model using a simple Bounce-Back (BB) boundary and the standard subgrid-scale (SGS) model; (2) a model incorporating the Wall Function (WF) and Suga's refined SGS TKE calculation; and (3) an integrated model combining the WF-Suga framework with the Lagrangian Stochastic Model (LSM) and a velocity limitation.The accuracy of these models was evaluated based on their ability to replicate the theoretical logarithmic profile of particle concentration in a neutral boundary layer:

$$\frac{C_0 - C}{C^*} = \frac{1}{\kappa} \ln \frac{z}{z_0} \tag{13}$$

where C_0 represents the particle density at ground level, C^* is the friction density, κ is the von Kármán constant, and z_0 is the roughness length.The results demonstrated that the WF-Suga-LSM model provided the highest correlation with the logarithmic profile. In contrast, models without the LSM or the specialized SGS TKE tended to deviate from the expected distribution, particularly in the near-wall region. Furthermore, the necessity of the speed limitation–defined as $|u_s| < k_{sgs}^{1/2}$–was confirmed through an analysis of the particle velocity variance. Without this limitation, the stochastic term in the LSM produced unrealistically high velocity fluctuations near the boundary. Implementing the speed limit successfully reduced this variance, ensuring a stable and physically consistent velocity distribution across the vertical profile.By adopting this validated WF-Suga-LSM framework with speed limitations, this research ensures that the fundamental transport mechanisms are correctly modeled before introducing the complexities of the AIJ Case H urban block.

2.2 Source Area Modeling for Scalars and Fluxes

In micrometeorological measurements, particularly within complex urban environments, it is critical to determine the spatial representativeness of data collected at a specific height. While traditional models often assume homogeneous surfaces, real-world applications require an understanding of which specific surface areas influence a sensor. Schmid (1994) addressed this by defining the Source Area as the surface region contributing to a measured scalar concentration or flux, using the concept of a Source Weight Function (SWF).

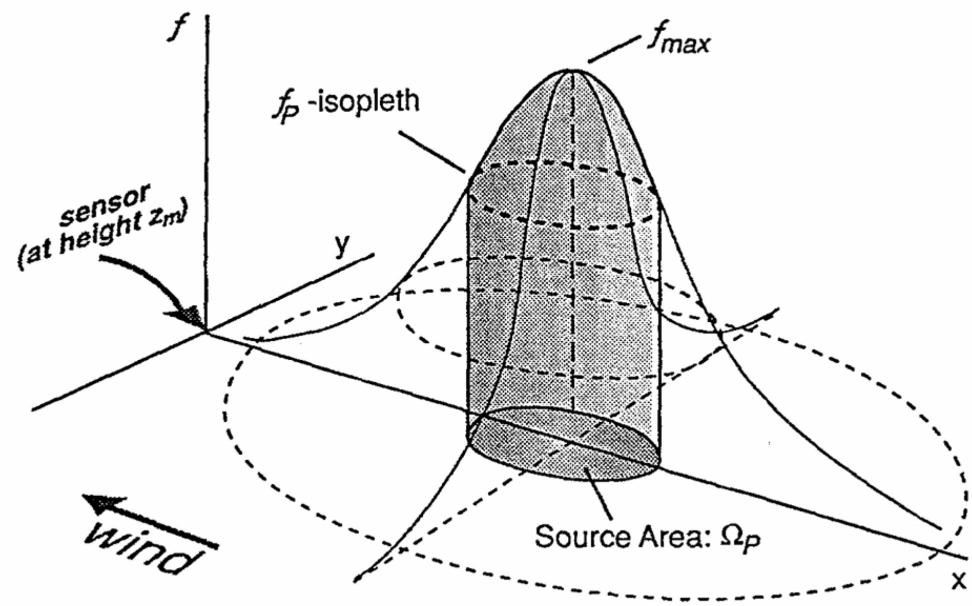


Figure 1: Conceptual illustration of the source area model, adapted from Schmid (1994). The 3D surface represents the source weight function, which shows how much each ground location influences the sensor. The influence is small near the sensor, reaches a peak at a certain upwind distance, and then falls off. The shaded region under the horizontal slice (isopleth f_P) represents the fraction P of the total influence. The source area Ω_P is the two-dimensional region on the x-y plane formed by the downward projection of this slice.

The relationship between a measurement at a specific point and the surface sources is fundamentally described by a convolution integral. Let $\eta(\mathbf{r})$ represent a measured quantity at position $\mathbf{r}$, and $Q(\mathbf{r}')$ represent the source strength at position $\mathbf{r}'$. The measurement is linked to the source via a probability transfer function, or Green's function, f:

$$\eta(\mathbf{r}) = \int_R Q(\mathbf{r}')f(\mathbf{r} - \mathbf{r}')d\mathbf{r}' \tag{14}$$

In this context, f represents the probability that material emitted at the surface location $\mathbf{r}'$ will arrive at the sensor location $\mathbf{r}$.

For measurements taken in the atmospheric surface layer at a height z_m, the general 3D convolution is reduced to a 2D surface integral. Here, z_0 represents the surface roughness length:

$$\eta(0, 0, z_m) = \int_{-\infty}^{\infty} \int_{-\infty}^{x} Q_\eta(x, y, z_0) f(-x, -y, z_m - z_0) dxdy \tag{15}$$

To represent a specific point source at coordinates (x_s, y_s, z_0) with strength Q_u, the source term is defined using Dirac delta distributions (δ):

$$Q(x, y, z_0) = Q_u \delta(x - x_s)\delta(y - y_s)\delta(z - z_0) \tag{16}$$

By substituting the point source definition into the integral, the relationship simplifies to a direct link between the source strength and the sensor through the source weight function:

$$\eta(0, 0, z_m) = Q_u f(x_s, y_s, z_m - z_0) \tag{17}$$

The "Source Area" is defined as the smallest surface area that accounts for a specified fraction P (e.g., 50%) of the total influence on the measured value. This is calculated by integrating the source weight function over the region where its values exceed a threshold f_P:

$$P = \frac{1}{\Phi_{tot}} \iint_{f(x,y) \geq f_P} f(x, y) dxdy \tag{18}$$

where the total influence Φ_{tot} is the integral of $f(x, y)$ over the entire surface:

$$\Phi_{tot} = \iint f(x, y) dxdy \tag{19}$$

2.3 Evolution of Flux Footprint Models

While analytical models provide a baseline for understanding source areas, modern environmental monitoring requires a balance between accuracy and computational speed. Kljun et al. (2015) highlights the current challenge in the field, noting:

> "Despite the widespread use of footprint models, the selection of a suitable model still poses a major challenge. Complex footprint models based on large-eddy simulation... can, to a certain degree, offer the ability to resolve complex flow structures... and surface heterogeneity. However, to date they are laborious to run, still highly CPU intensive... and hence can in practice be applied only for case studies over selected hours or days."

As described by Kljun et al. (2015), there is a clear trade-off in the study of atmospheric dispersion:

- **Complex Models:** These include the Large-Eddy Simulations (LES) and Lagrangian Stochastic Particle Dispersion (LPD) models. While they can handle complex urban geometry like street canyons, they require significant supercomputing power and are difficult to use for long-term data.
- **Analytical Models:** Simplified models like those by Schmid (1994) are fast and efficient but are often limited to specific atmospheric conditions and heights.

In this thesis, the use of the GPGPU-LBM solver on the TSUBAME 4.0 supercomputer aims to bridge this gap. By leveraging GPU acceleration, the simulation can perform the ”laborious” LES and particle tracking calculations at a much faster rate than traditional CPU-based models. This allows the research to evaluate if the high-fidelity results of a complex simulation align with the simplified footprint estimates often used in environmental monitoring.

3 Methodology

3.1 Validation and Numerical Workflow

3.1.1 Benchmark Case: AIJ Case H

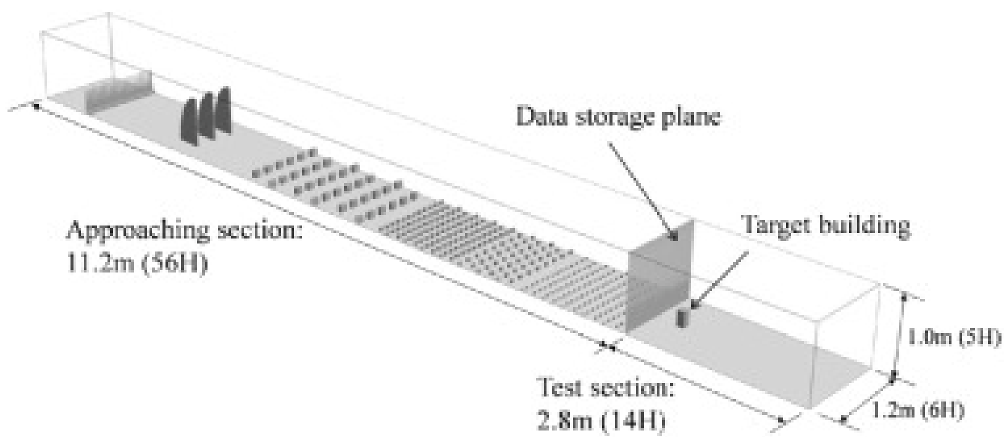


Figure 2: Experimental setup for AIJ Case H at Tokyo Polytechnic University,isolated building model and the arrangement of spires and roughness elements in the approach section to induce a turbulent boundary layer (adapted from Architectural Institute of Japan, 2021; Okaze et al., 2021).

The numerical validation of the Lattice Boltzmann Method (LBM) in this study is based on Case H of the Architectural Institute of Japan (AIJ) benchmark, which involves an isolated building model within a turbulent boundary layer. The physical experiment was conducted at Tokyo Polytechnic University using a 1:250 scale model. The building is a rectangular prism with a height-to-width ratio of $1 : 1 : 2$ ($H = 0.20$ m).The wind tunnel test section measures $2.8\,\text{m} \times 1.2\,\text{m} \times 1.0\,\text{m}$, corresponding to normalized dimensions of $14H \times 6H \times 5H$. Turbulence in the approach section was induced using spires and roughness elements to simulate a realistic atmospheric boundary layer. For the pollutant dispersion study, a tracer gas release point was located 50 mm ($0.25H$) from the leeward side of the building. To ensure comparability across different scales, the reference velocity U_H is defined as the mean inflow velocity at the building height.

The LBM results in this research are cross-validated against the comprehensive Large Eddy Simulation (LES) study conducted by Okaze et al. (2021). Their study utilized OpenFOAM to investigate the influence of various subgrid-scale (SGS) models—including the Standard Smagorinsky (S), Dynamic Smagorinsky (DS), Wall-Adapting Local Eddy-Viscosity (WALE), and Coherent Structure (CS) models—on turbulent statistics. Notably, Okaze et al. (2021) evaluated grid sensitivity across 10, 20, and 40-grid resolutions relative to the building width. This thesis focuses on comparing LBM performance against the validation study, using the same validation standards defined in the paper, which is a hit rate (q) of ≥ 0.66 and a factor of two observations (FAC2) of > 0.5. By comparing the LBM-LSM results directly to these established OpenFOAM benchmarks, the predictive accuracy of the model can be evaluated.

A significant factor in the accuracy of the benchmark study was the choice of the computational grid. Okaze et al. (2021) utilized a domain of $12.5H(x) \times 6.0H(y) \times 5.0H(z)$ and intentionally avoided non-conformal grids—setups where the grid is made finer or coarser in specific areas to save on calculation time. They observed that sudden changes in grid size can cause mathematical errors and oscillations (unstable values) at the boundaries where the different grid scales meet. To ensure their comparisons of the physics models remained pure, they used a consistent, orthogonal grid system throughout their study.

In this research, the LBM-LES solver employs a uniform grid that remains perfectly even across the entire simulation space. This grid is technically non-conformal to the building's geometry, but do not bend or wrap to match the building's surface so it avoids the errors mentioned by Okaze et al. (2021). Because every grid cell in the LBM simulation is the same size regardless of its location or proximity to the building, the wind field calculation maintains a high degree of mathematical stability.

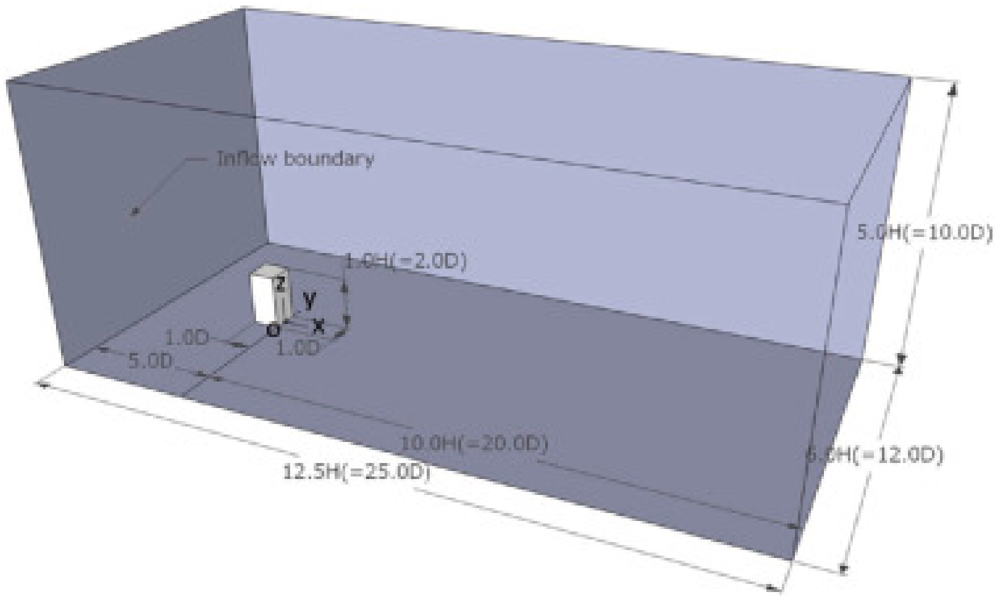


Figure 3: Numerical domain configuration and measurement locations for the LES validation study (adapted from Okaze et al., 2021).

3.1.2 Computational Implementation and Workflow

The simulation is designed to run efficiently by using the Graphics Processing Units (GPU) to handle the heavy mathematical workload. The source code is written using the CUDA platform, which allows a standard programming language like C to run thousands of calculations simultaneously on the GPU hardware (NVIDIA Corporation, 2024). The software is organized into modules where the wind simulation (LBM) and the particle tracking (LPT) are connected. The wind field is calculated first, and that information is then used to move the particles. The system is managed by configuration files (ending in `.mak`) and is organized into six functional parts:

1. **Simulation Setup and Constants**: This part handles the initial settings, such as the size of the wind domain and the physical properties of the air and particles. It uses header files (like `Define_user.h`) to set the desired variables for the specific research case before the calculation begins.

2. **Data Memory Management**: Before calculations start, information must be moved from the computer's main memory to the GPU's high-speed memory. This module ensures that the wind grid and particle positions are stored where the GPU can access them quickly.

3. **Wind Field Calculation (LBM)**: This is the core of the wind solver. It calculates how the air moves and collide on a grid. Because these collisions happen at every point in space at the same time, the

GPU handles this by calculating every grid cell in parallel.

4. **Boundary and Obstacle Handling**: This module manages how the wind interacts with boundaries, such as the ground surface or the building model. It applies the specific roughness settings (z_0) and ensures the wind cannot pass through solid walls.

5. **Particle Movement (LPT)**: Once the wind speed is known at a specific point, this module calculates the new position of each individual particle. It accounts for the forces the wind exerts on the particles, moving them through the complex flow patterns around the building.

6. **Data Output and Recording**: At set intervals, the simulation saves the state of the wind and particles to the storage system. These are the files (such as the `.csv` and binary results) used for the statistical validation and profile comparisons discussed in the results section.

The graphical overview of the system architecure described above can be seen in Figure 4.

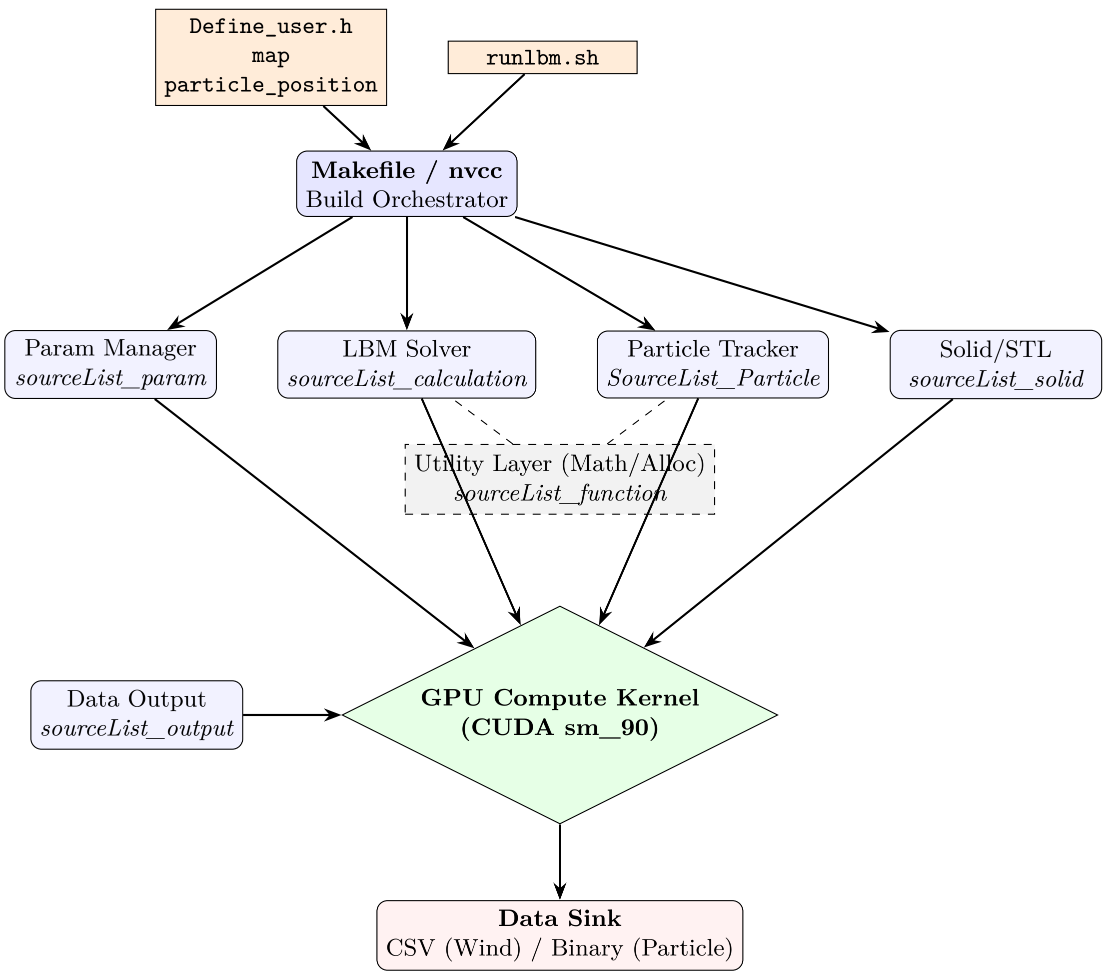


Figure 4: System architecture of the LBM simulation framework

The same source code for the simulation can also be used when running the simulation on a larger scale. However, to handle an even larger calculation and longer periods of time, multiple nodes and GPUs must be used. As such, even though the base code and the output process is the same, the code is ran in the

TSUBAME 4.0 Supercomputer. A shell script `run_program.sh` is used to queue the LBM-LSM simulation program into the supercomputer job scheduler, before the simulation is run. The system architecture diagram of this process can be viewed in Figure 5.

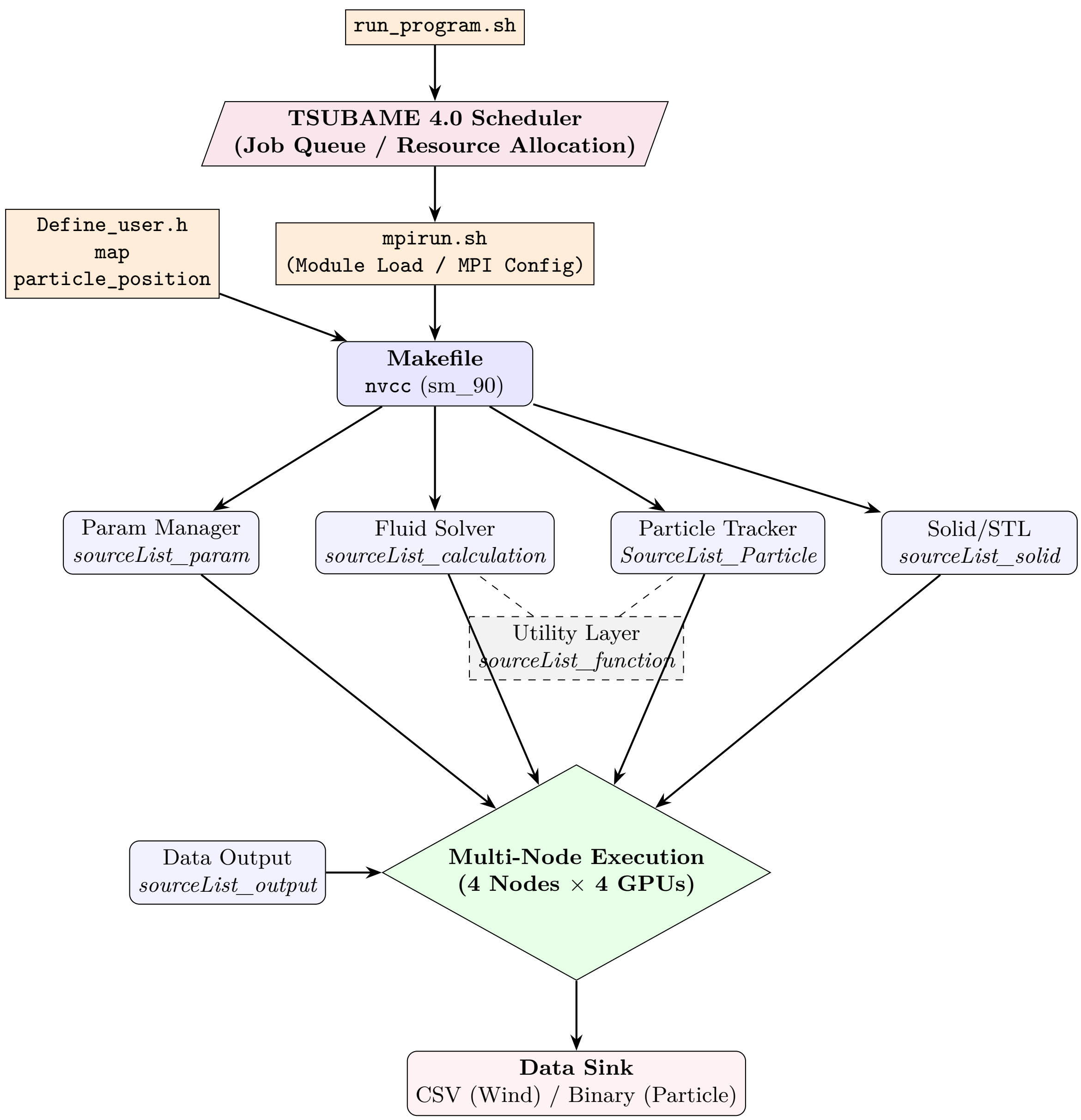


Figure 5: System architecture of the LBM simulation framework executed using the TSUBAME 4.0 Supercomputer

3.2 Simulation Parameters and Case Setup

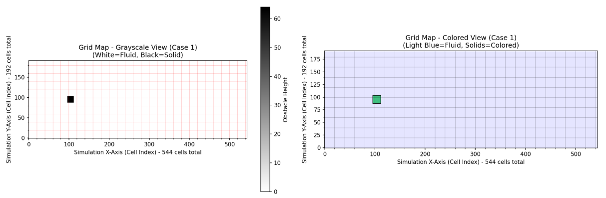
The specific configurations for the validation and research cases are summarized in Table 1. These cases represent the transition from local prototyping to large-scale supercomputer execution.

Additionally, to provide a spatial context for the validation studies, Figure 6 presents the plan-view (birds-eye view) layouts for the various simulation domains. These maps illustrate the placement of the

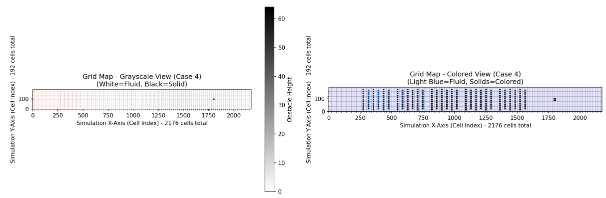
Table 1: Detailed Simulation Case Parameters

Case	**Environment**	**Domain Size (x, y, z in m)**	**Duration (s)**	**Approach**	H_{cube} **(m)**	z_0 **(m)**	U_0 **(m/s)**	**Flow Rate**
1	Local Server	$832 \times 384 \times 240$	2400	No	–	0.001	6.0	150 p/s
2	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	8	0.001	6.0	150 p/s
3	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	16	0.001	6.0	150 p/s
4	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	20	0.001	6.0	150 p/s
5	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	24	0.001	6.0	150 p/s
6	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	32	0.001	6.0	150 p/s
7	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	16	0.1	6.0	150 p/s
8	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	20	0.1	6.0	150 p/s

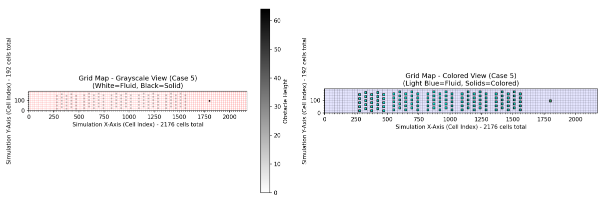
central building model relative to the upwind approach sections.



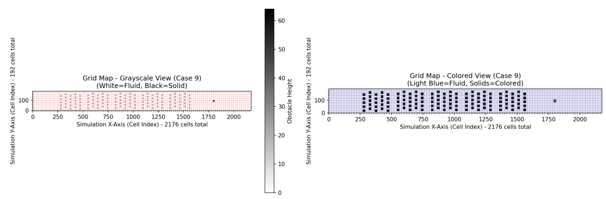
(a) Case 1: No approach flow



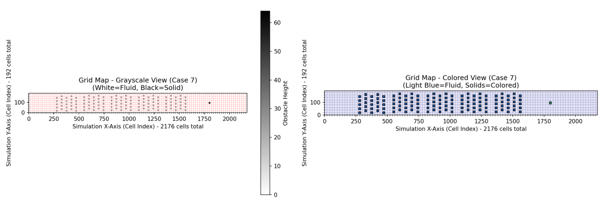
(b) Case 2: 8m blocks



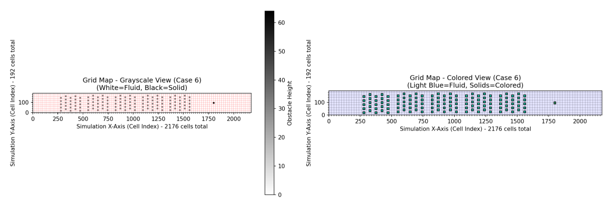
(c) Cases 3 & 7: 16m blocks



(d) Cases 4 & 8: 20m blocks



(e) Case 5: 24m blocks



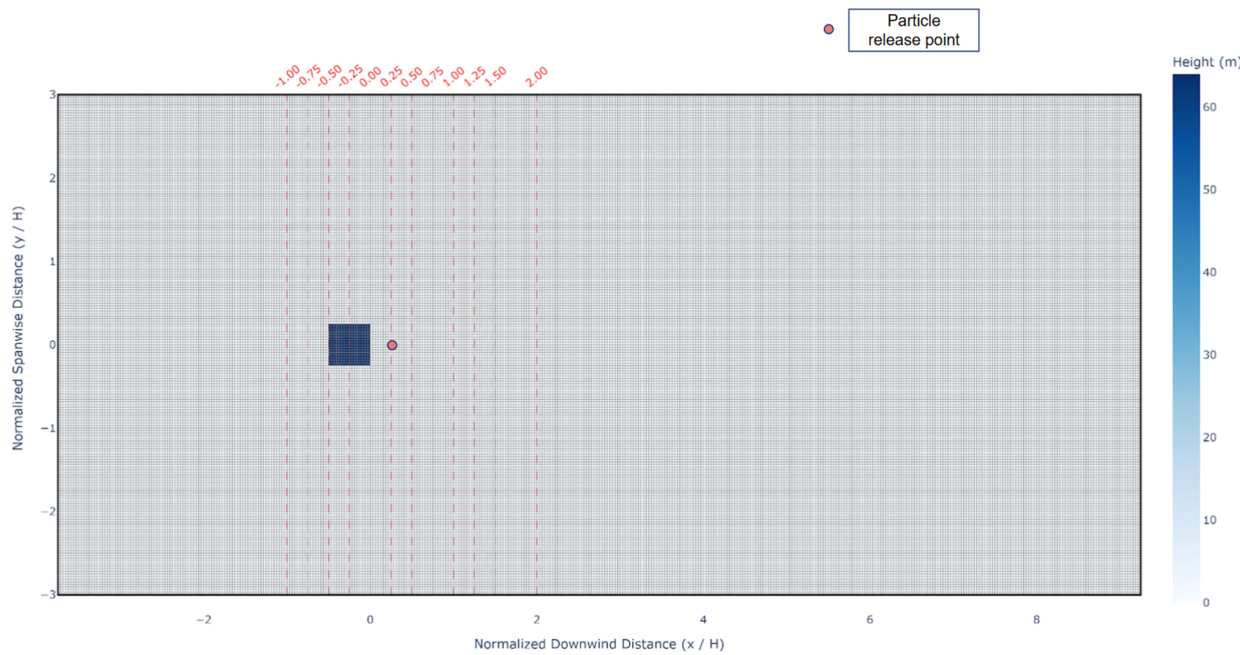
(f) Case 6: 32m blocks

Figure 6: Plan-view visualizations of the computational domains for all simulation cases. Cases 7 and 8 utilize the same geometric configurations as Cases 3 and 4, respectively, but feature an increased surface roughness ($z\_0 = 0.1$ m).

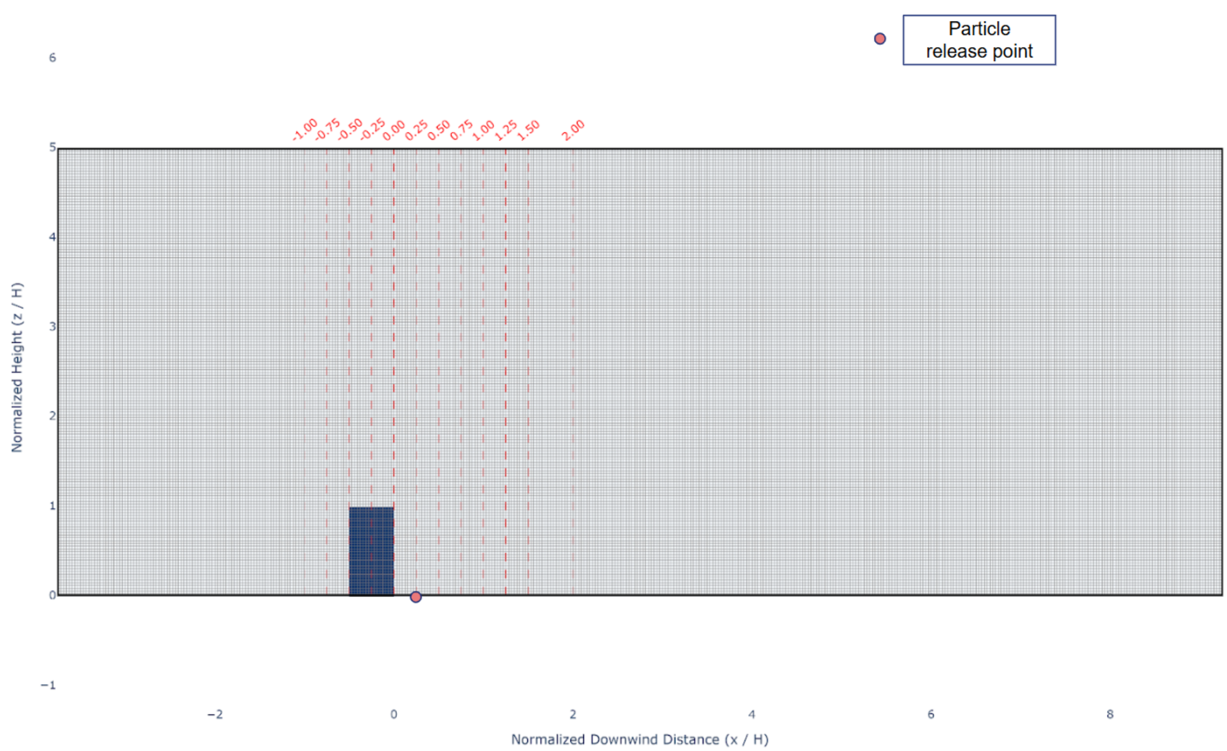
The domain layouts demonstrate the transition from the isolated building simulation in Case 1 to the extended domains used on the TSUBAME 4.0 supercomputer. The extended upwind area allows for the physical development of a turbulent boundary layer as the wind passes over the array of roughness cubes. As the height of these blocks (H_{cube}) increases across the cases, the resulting turbulence intensity is expected to better match the conditions of the AIJ Case H wind tunnel experiment.

To ensure a direct comparison with the AIJ wind tunnel benchmarks, statistical data were extracted at specific coordinates within the LBM domain. As shown in Figure 7, 12 vertical profiles were selected along the center-line of the domain. These locations cover the windward, rooftop, and leeward wake regions, defined by their normalized distance from the building (x/H).

The xy-plane view (Figure 7a) confirms the streamwise alignment of the points, while the xz-plane view



(a) Bird's-eye view (xy-plane)



(b) Spanwise view (xz-plane)

Figure 7: Spatial distribution of measurement points within the LBM domain, showing the alignment with the AIJ experimental case.

(Figure 7b) illustrates the vertical extent of the profiles relative to the building height H. This setup allows for the systematic evaluation of the flow field and pollutant dispersion in the near-field region.

3.3 Physical and Dimensionless Parameters

To ensure the dynamic similarity between the wind tunnel experiment and the Lattice Boltzmann Method (LBM) simulation, the Reynolds number (Re) is calculated. This dimensionless value represents the ratio of inertial forces to viscous forces.

Reynolds Number (Re) For this study, the characteristic length is defined as the building height H, and the velocity is the wind speed at that height U_H:

$$Re = \frac{U_H \cdot H}{\nu} \tag{20}$$

While the experimental Re_{exp} is 4.3×10^4, the LBM simulation on TSUBAME 4.0 operates at a significantly higher effective Reynolds number ($Re_{LBM} \approx 2.09 \times 10^7$). This discrepancy is strategically justified by the physical principles of urban flow modeling. According to the guidelines established by Snyder (1981), a Reynolds number of 1.1×10^4 serves as a conservative threshold for ensuring that the flow characteristics around cube-like models are independent of the specific Re value.

Since both the experiment and the simulation significantly exceed this critical threshold, the large-scale turbulent structures that serve as the primary focus of the Large Eddy Simulation (LES) approach remain physically consistent. The higher Re_{LBM} utilized in this thesis reflects the high-resolution grid permitted by the supercomputing environment, allowing for a more detailed capture of the flow without violating the principle of Reynolds-number independence.

3.3.1 Wind and Turbulence Data Post-Processing and Statistical Analysis

The Large Eddy Simulation (LES) results were post-processed to extract time-averaged wind statistics and turbulence characteristics. The simulation domain was decomposed into four parallel sub-domains (indexed 0000 to 0003) for computational efficiency on the TSUBAME 4.0 supercomputer.

Data analysis focused primarily on the fourth sub-domain (0003), which contains the building model and the validation points of interest, excluding the approach flow sections (0000–0002).

Data Structure and Extraction Wind field data were exported as planar cross-sections (xy-planes) at various vertical grid heights (k). The raw output files follow a stacked structure where each height level is preceded by a header indicating the vertical grid index, followed by a 192×544 matrix representing the spanwise (y) and streamwise (x) velocity components. On the small domain simulation (Case 1), the data structure is similar, except for the inexistence of the approach section. Thus, for the small domain simulation, only one sub-domain, 0000, exists, and the analysis is done over that sub-domain only.

The analysis utilizes time-averaged velocity components ($\bar{u}, \bar{v}, \bar{w}$) and second-order velocity moments ($\overline{u^2}, \overline{v^2}, \overline{w^2}$), averaged over 540,000 time steps (equivalent to 5,400 physical seconds) to ensure statistical convergence. These variables correspond to the output files `xy_um`, `xy_vm`, `xy_wm` (first-order moments) and `xy_uu`, `xy_vv`, `xy_ww` (second-order moments).

Turbulence Statistics

To evaluate the turbulent flow characteristics, Reynolds stresses were calculated by subtracting the square of the mean velocity from the mean of the squared velocity. The streamwise ($\langle u'^2 \rangle$), spanwise ($\langle v'^2 \rangle$), and vertical ($\langle w'^2 \rangle$) Reynolds normal stresses were derived as follows:

$$\langle u'^2 \rangle = \overline{u^2} - (\bar{u})^2 \tag{21}$$

$$\langle v'^2 \rangle = \overline{v^2} - (\bar{v})^2 \tag{22}$$

$$\langle w'^2 \rangle = \overline{w^2} - (\bar{w})^2 \tag{23}$$

Root Mean Square (RMS) Fluctuations

The Root Mean Square (RMS) velocity fluctuations, which indicate the intensity of turbulence in each direction, were calculated and normalized by the reference approach flow velocity (U_H) at building height H:

$$u_{rms} = \frac{\sqrt{\langle u'^2 \rangle}}{U_H}, \quad v_{rms} = \frac{\sqrt{\langle v'^2 \rangle}}{U_H}, \quad w_{rms} = \frac{\sqrt{\langle w'^2 \rangle}}{U_H} \tag{24}$$

Turbulence Kinetic Energy (TKE)

The Turbulence Kinetic Energy (k), representing the mean kinetic energy per unit mass associated with eddies in the turbulent flow, was computed as half the sum of the Reynolds normal stresses:

$$k = \frac{1}{2} \left(\langle u'^2 \rangle + \langle v'^2 \rangle + \langle w'^2 \rangle \right) \tag{25}$$

To handle numerical stability during post-processing, a constraint was applied such that any negative stress values resulting from floating-point errors were converted to zero before the square root operation ($\max(0, \sigma^2)$). All statistical results were mapped to the physical domain coordinates, and normalized with height H as well as approach streamwise wind speed U_H to facilitate direct comparison with the Architectural Institute of Japan (AIJ) wind tunnel experimental benchmarks.

3.3.2 Lagrangian Particle Data Post-Processing and Statistical Analysis

The raw simulation output is generated in a condensed binary format to manage the high volume of data produced by the Lagrangian tracking. These binary files are recorded at 1-second intervals (equivalent to every 100 simulation time steps). For each output interval, a set of specialized files is created, including `header`, `index`, `position`, `velocity`, and `SGS_velocity`. Each file contains the specific physical attributes of all particles active within the domain at that specific moment. For instance, a file named `position0-001.bin` stores the three-dimensional Cartesian coordinates (x, y, z) of every particle at $t = 1$ second.

To translate these large-scale datasets into physically interpretable results, a dedicated analysis suite was developed in C++. This suite utilizes a modular design where specific physical quantities are calculated by independent components managed by a central execution program. The workflow is handled by a C++ executable, compiled using a `Makefile` that links distinct modules for density calculation, flux analysis, vertical profiling, and Flux Footprint Prediction (FFP).

The system is designed for high flexibility across different research scenarios. Rather than fixed internal parameters, the tool employs a command-line interface where the domain dimensions, grid resolution, time steps, and output intervals are provided as arguments at the time of execution. This allows the same analytical core to be applied to various cases—ranging from local isolated building tests to large-scale urban simulations—without requiring the code to be re-compiled.

The analysis is typically managed through batch scripts (`run.bat`), allowing the framework to simultaneously perform multiple tasks, such as extracting particle density profiles at specified vertical heights (z_{out})

and computing source area contributions for defined sensor locations.

To process the millions of Lagrangian trajectories efficiently, the post-processor utilizes OpenMP for multi-threading (`OMP_NUM_THREADS`). This allows the software to use multiple CPU cores simultaneously to read the binary data and perform calculations in parallel. The final results are aggregated into averaged statistical summaries in `.csv` or text format, which are then used for final visualization and plotting in Python.

Normalized Concentration (C_{norm})

To allow for a direct comparison between experimental gas data and the Lagrangian particle model, concentration values are normalized based on the flow rate Q, velocity U_H, and building scale H.

$$C_{norm} = \frac{C_o \cdot U_H \cdot H^2}{Q} \tag{26}$$

3.4 Statistical Validation Metrics

Following the AIJ guidelines for urban wind environments, four metrics are used to compare the predicted values (P_i) against the observed experimental data (O_i).**Hit Rate (q)**The Hit Rate measures the reliability of the model. A data point is a "hit" if the prediction is within a relative tolerance D (usually 0.25) or an absolute tolerance W (usually 0.05).

$$q = \frac{1}{N}\sum_{i=1}^{N} n_i, \quad n_i = \begin{cases} 1 & \text{if } \left|\frac{P_i - O_i}{O_i}\right| \le D \text{ or } |P_i - O_i| \le W \\ 0 & \text{else} \end{cases} \tag{27}$$

Factor of Two (FAC2) FAC2 is a robust measure of the overall spread. It checks if the predicted value is between half and double the observed value.

$$\text{FAC2} = \frac{1}{N}\sum_{i=1}^{N} n_i, \quad n_i = \begin{cases} 1 & \text{if } 0.5 \le \frac{P_i}{O_i} \le 2.0 \\ 0 & \text{else} \end{cases} \tag{28}$$

Error Metrics (MAE and RMSE) The Mean Absolute Error (MAE) provides the average magnitude of the errors, while the Root Mean Square Error (RMSE) gives a higher weight to large outliers, providing a stricter assessment of simulation accuracy.

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} |P_i - O_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(P_i - O_i)^2} \tag{29}$$

4 Results and Discussion

4.1 Wind and Turbulence Statistics

This section will explore the results of the wind and turbulence produced by the LBM-LES simulation, as detailed in section 3, from the comparison of the vertical profile of the streamwise velocity (u), spanwise velocity (v), vertical velocity (w), streamwise velocity fluctuation (u_{rms}), spanwise velocity fluctuation (v_{rms}), vertical velocity fluctuation (w_{rms}), as well as turbulent kinetic energy (TKE).

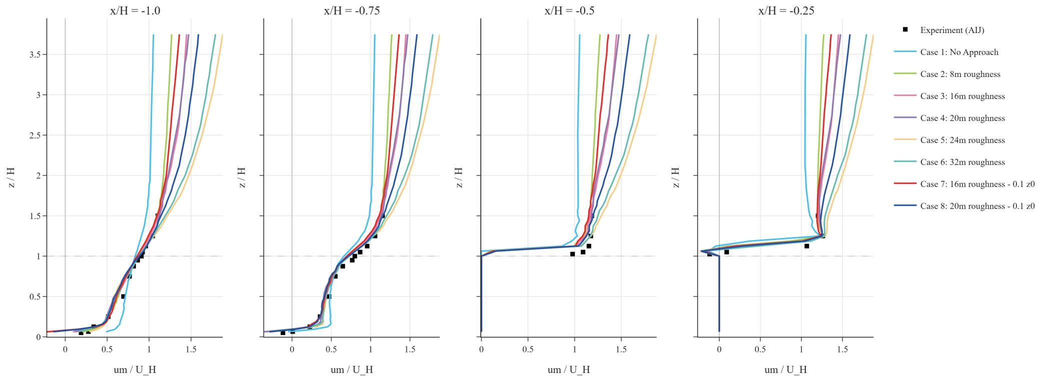
4.1.1 Wind and Turbulence Profile

Figure 8 illustrates the vertical profiles of the normalized streamwise velocity (u/U_H) at the locations defined in Figure 7. A primary observation is that the simulation's ability to reproduce the experimental data improves significantly as the inflow turbulence increases, driven by the added roughness in the approach section. Even in the most simplified cases lacking approach roughness, the LBM results maintain the general characteristic shape observed in the AIJ wind tunnel experiments, capturing the fundamental flow deceleration and wake recovery. However, the introduction of even minimal roughness provides a substantial increase in accuracy, effectively closing the gap between the numerical and experimental data. Beyond this initial improvement, the profiles exhibit a plateau in refinement; the variation between different high-roughness cases is minimal, suggesting that once the atmospheric boundary layer is sufficiently developed, the broad-scale structure of the streamwise velocity becomes stable. This indicates that the presence of upstream turbulence is more critical for profile accuracy than the specific magnitude of the roughness height once a certain threshold is reached.

Figure 9 presents the vertical profiles of the normalized spanwise velocity (v/U_H). In contrast to the streamwise results, the spanwise profiles are largely inconclusive regarding the impact of approach roughness. Both the experimental data and numerical results exhibit small-scale fluctuations with magnitudes on the order of 10^{-2}, representing minor lateral shifts in the wind direction. Due to the stochastic nature of these lateral movements and their negligible magnitude relative to the primary flow, the patterns appear relatively random. Consequently, no significant correlation can be established between the increase in upstream turbulence and the predictive accuracy of the spanwise velocity component in the near-field wake.

Figure 10 illustrates the vertical profiles of the normalized vertical wind speed (w/U_H). These results provide the most significant insights into the model's performance, as they reveal a clear sensitivity to the approach flow conditions. At the initial inflow location ($x/H = -1.0$), there is a general underprediction of the vertical velocity. However, as the flow reaches $x/H = -0.75$, the profiles across all cases show relatively good agreement with the experimental data. A critical discrepancy emerges at the building's center-line ($x/H = -0.25$), where the case lacking approach roughness significantly overpredicts the upward vertical velocity. This overprediction becomes more pronounced in the leeward wake regions (behind the building), where the negative vertical wind speed—representing the downwash in the recirculation zone—is overestimated by a significant margin. This bias persists in the lower-roughness cases, albeit to a lesser degree, but is relatively well-mitigated in Cases 5 through 8. Notably, these successful cases either employ larger roughness blocks (24 m and 32 m in Cases 5 and 6) or combine moderate blocks (16 m and 20 m) with an increased surface roughness parameter ($z_0 = 0.1$ in Cases 7 and 8). This suggests that a specific threshold of turbulent intensity and surface friction is required to correctly model the vertical momentum exchange in the building's near-field.

Figure 11 illustrates the vertical profiles of the root-mean-square (RMS) streamwise velocity fluctuations



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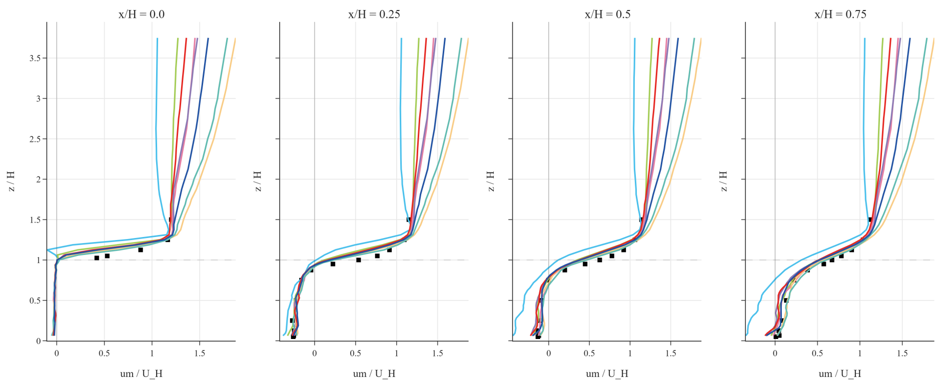

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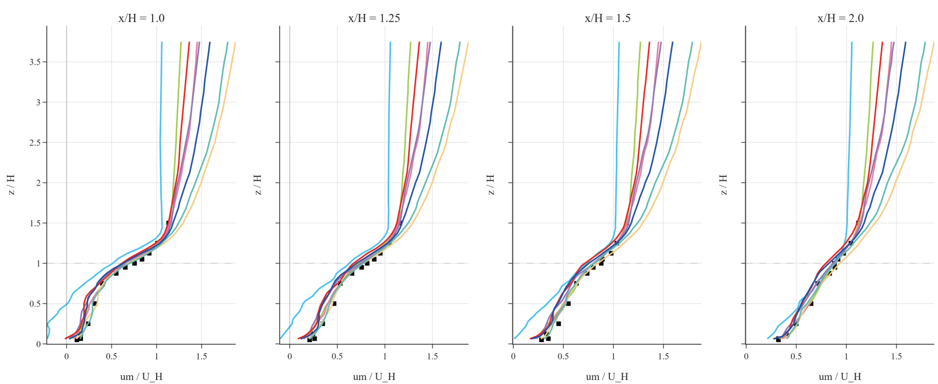
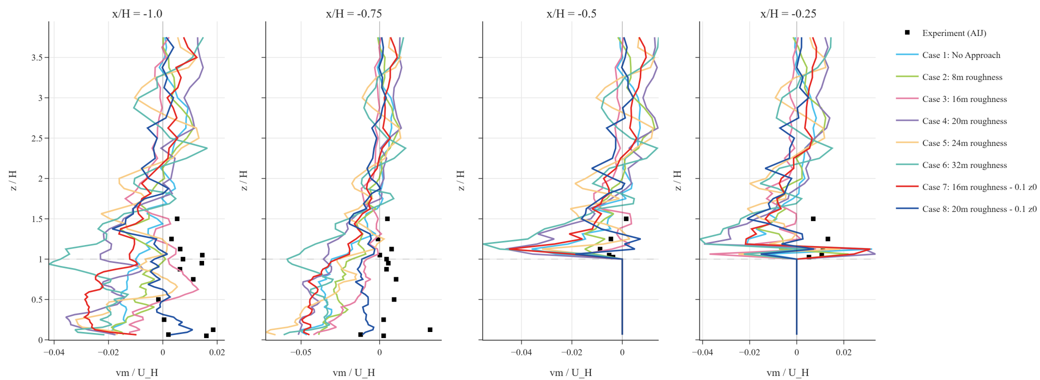


(c)

Figure 8: Normalized mean streamwise velocity ($\bar{u}/U_H$) profiles at (a) Windward locations ($x/H = -1.0$ to -0.25), (b) Near Wake locations ($x/H = 0.0$ to 0.75), and (c) Far Wake locations ($x/H = 1.0$ to 2.0).



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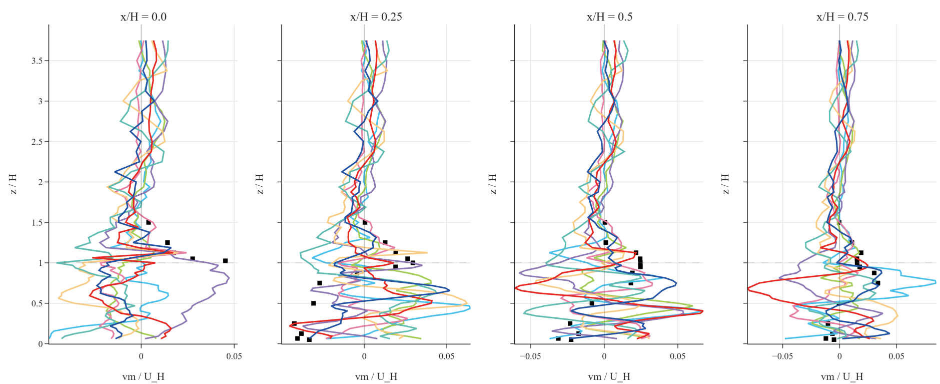

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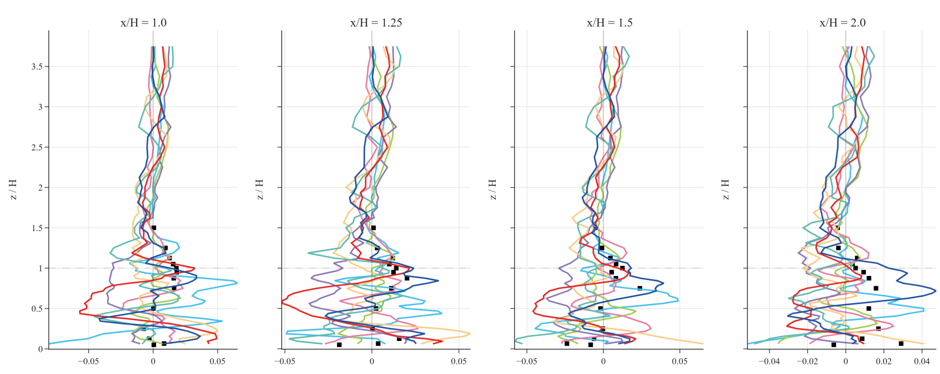
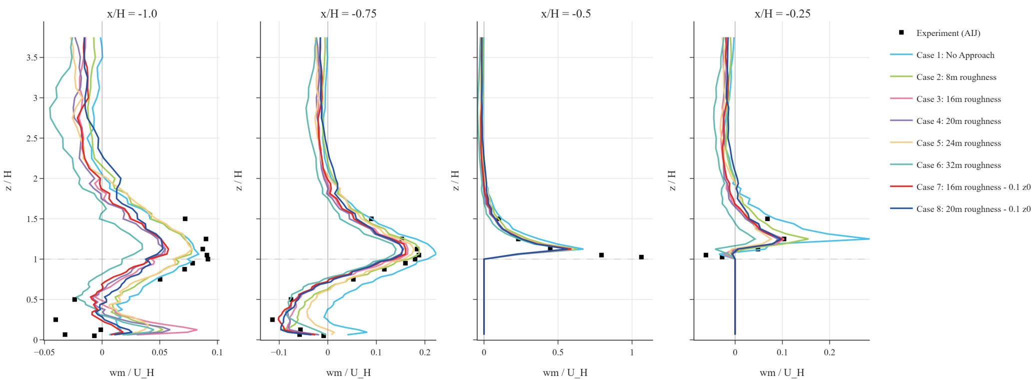


(c)

Figure 9: Normalized mean spanwise velocity ($\bar{v}/U_H$) profiles at (a) $x/H = -1.0$ to -0.25, (b) $x/H = 0.0$ to 0.75, and (c) $x/H = 1.0$ to 2.0.



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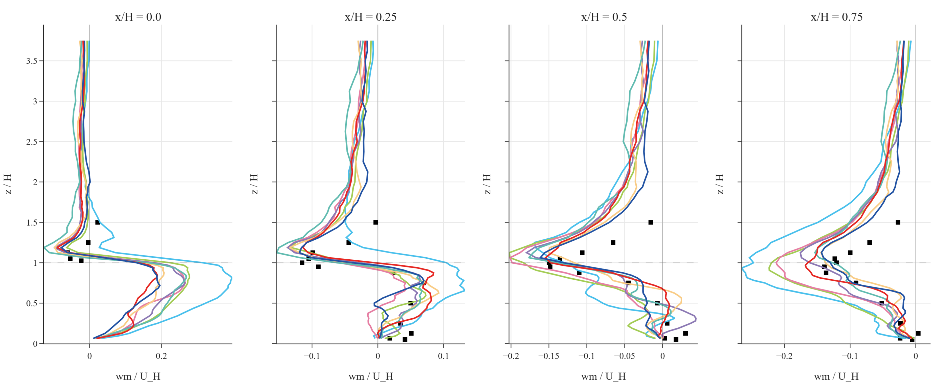

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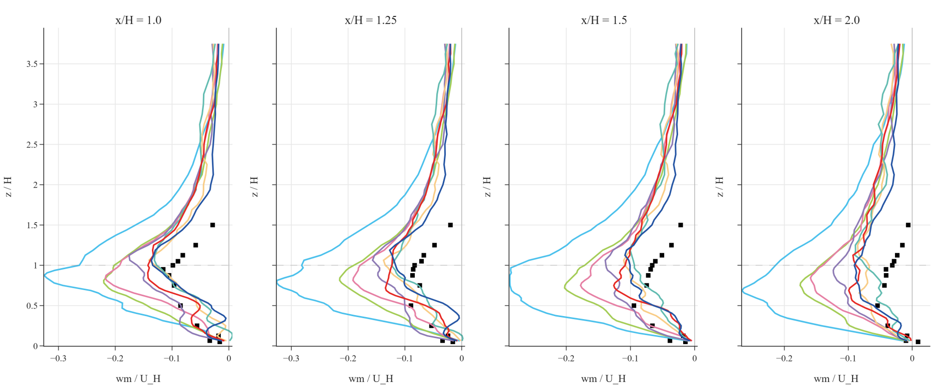


(c)

Figure 10: Normalized mean spanwise velocity ($\bar{w}/U_H$) profiles at (a) $x/H = -1.0$ to -0.25, (b) $x/H = 0.0$ to 0.75, and (c) $x/H = 1.0$ to 2.0.

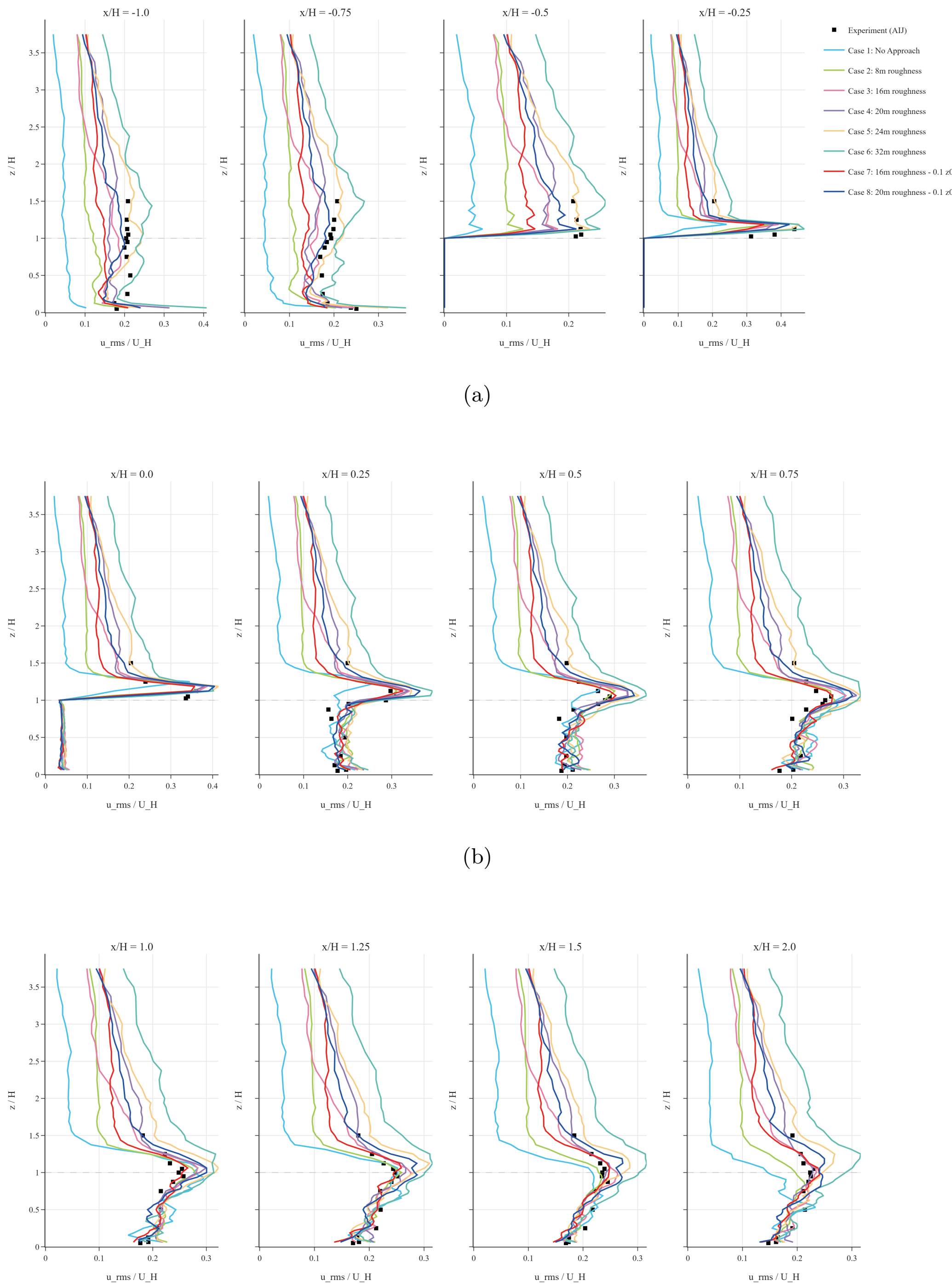


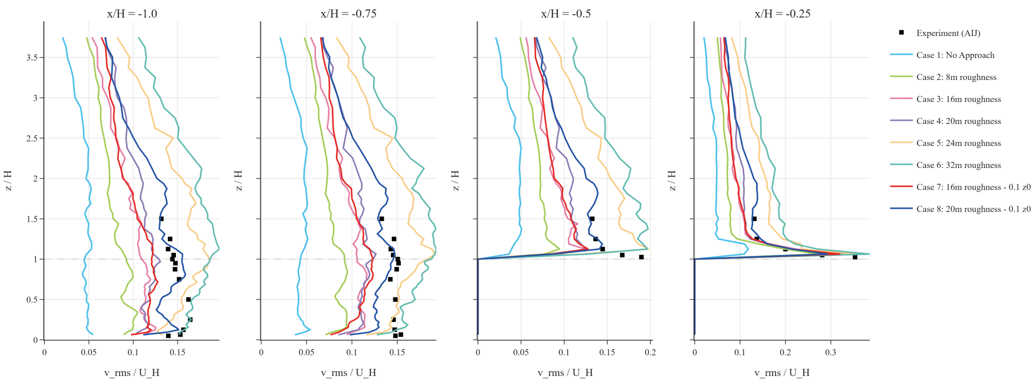
Figure 11: Normalized streamwise velocity fluctuations (u_{rms}/U_H) at (a) Upwind/Rooftop, (b) Leeward Side ($x/H = 0$ to 0.75), and (c) Downwind Wake ($x/H = 1.0$ to 2.0).

(u_{rms}/U_H). Unlike the mean velocity profiles, which showed general agreement across most cases, the u_{rms} results exhibit significant variability depending on the approach roughness. In the windward approach region ($x/H = -1.0$), the velocity variation is notably underpredicted, particularly in the case lacking an approach section. Accuracy improves as the roughness block size increases; however, the largest blocks (Cases 5 and 6) lead to a slight overprediction of the turbulence intensity. Directly above the building, the high-roughness cases and the medium-roughness cases with increased z_0 provide the most accurate predictions, while the case with no roughness severely underpredicts the fluctuations. In the immediate leeward wake, all cases capture the general pattern of the profile. Small roughness cases tend toward slight underprediction, whereas larger roughness blocks result in a slight overprediction. The discrepancy becomes most prominent in the far-wake region. Here, the no-roughness case fails to maintain the turbulent energy, leading to a severe underprediction of the profile. Conversely, while higher roughness slightly overestimates the variation, Cases 3 and 4 (medium roughness) and Cases 7 and 8 (medium roughness with increased z_0) demonstrate the highest overall predictive accuracy. This suggests that moderate turbulence levels best replicate the balanced energy state observed in the wind tunnel experiments.

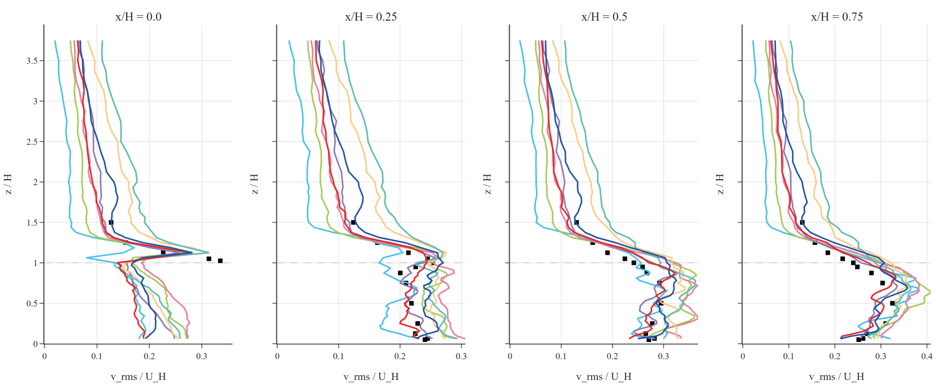
Figure 12 presents the vertical profiles of the normalized spanwise velocity fluctuations (v_{rms}/U_H). Similar to the observations in the u_{rms} results, the approach section without roughness severely underpredicts the spanwise turbulence levels. Conversely, the cases with the largest roughness blocks (Cases 5 and 6) lead to an overprediction of these fluctuations in the windward region. Interestingly, while the high-roughness cases provide the most accurate predictions directly above the building, this accuracy transitions into a magnitude overprediction as the flow enters the leeward wake. A notable phenomenon occurs in the no-roughness case behind the building: rather than a simple underprediction, the model produces a significant overprediction of v_{rms} with a distinct change in the profile shape. In these instances, the variation is characterized by a sharp peak reaching at height 0.5 H, whereas the experimental data and the more accurate simulation cases show a smoother, more stable curvature. While the large roughness blocks also overpredict the magnitude in the wake, they maintain the correct physical shape of the profile. Consistent with the results of other velocity components, the medium-roughness cases (Cases 3 and 4) and those combining medium roughness with increased surface friction (Cases 7 and 8) demonstrate the most reliable results overall. These cases avoid both the numerical instability seen in the no-roughness peaks and the excessive energy associated with the largest roughness obstacles.

Figure 13 illustrates the vertical profiles of the normalized vertical velocity fluctuations (w_{rms}/U_H). The results closely mirror the patterns observed in the spanwise fluctuations (v_{rms}), both in terms of magnitude and profile shape. A recurring phenomenon is observed in the case without an approach section, which exhibits a sharp, non-physical peak in the leeward wake. This sharp peak matches the position and anomalous shape seen in the v_{rms} results, further suggesting that the absence of upstream turbulence leads to numerical instabilities or an unnatural vortex structure behind the building. For the remaining cases, the patterns of underprediction in low-roughness setups and overprediction in high-roughness setups remain consistent. The profile shapes are nearly identical to those of the v_{rms} cases, with Cases 7 and 8 again providing the most stable and accurate fit to the AIJ wind tunnel data. The similarity between the v_{rms} and w_{rms} profiles indicates that the turbulence in the wake is largely isotropic in its fluctuations, and that the approach flow roughness is the primary factor in correctly capturing this vertical energy distribution.

Figure 14 presents the vertical profiles of the normalized Turbulent Kinetic Energy (TKE). As expected, the TKE profiles closely follow the trends observed in the v_{rms} and w_{rms} results, given that TKE is mathematically defined as half the sum of the variances of the velocity components, as defined in Equation 25.



(a)



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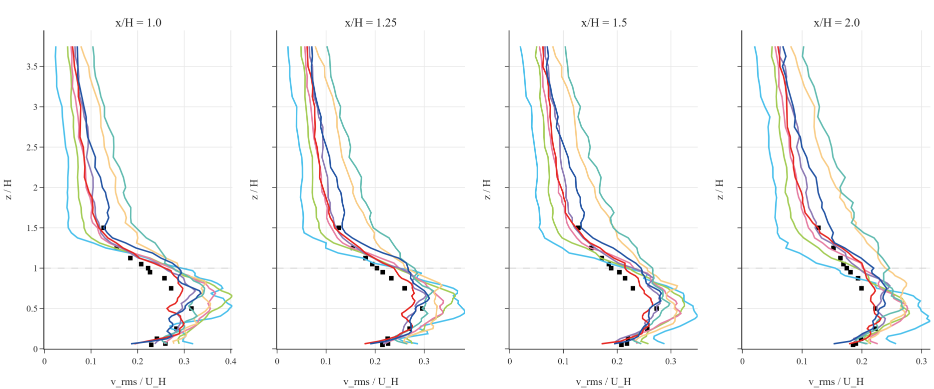
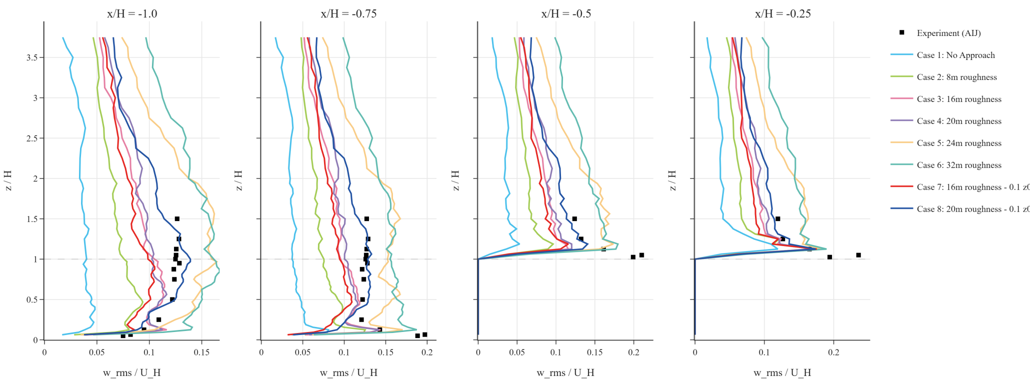
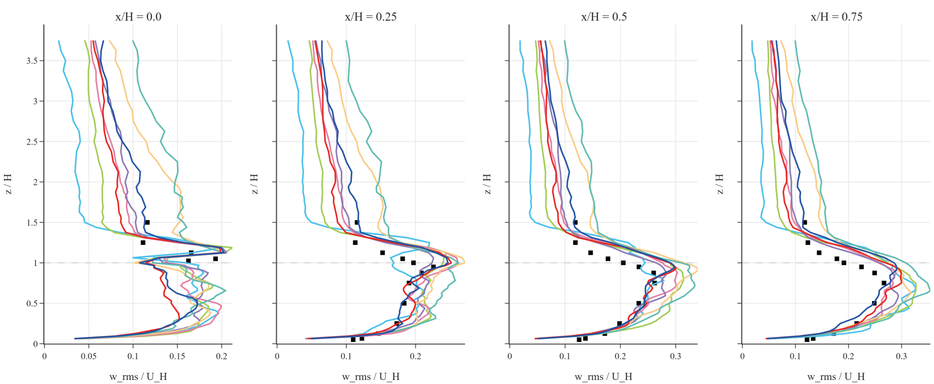


(c)

Figure 12: Normalized spanwise velocity fluctuations (v_{rms}/U_H) at (a) Upwind/Rooftop, (b) Leeward Side ($x/H = 0$ to 0.75), and (c) Downwind Wake ($x/H = 1.0$ to 2.0).



(a)



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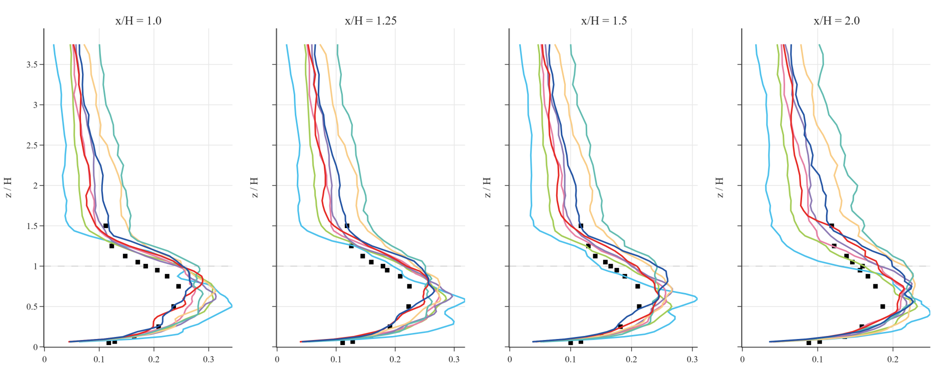
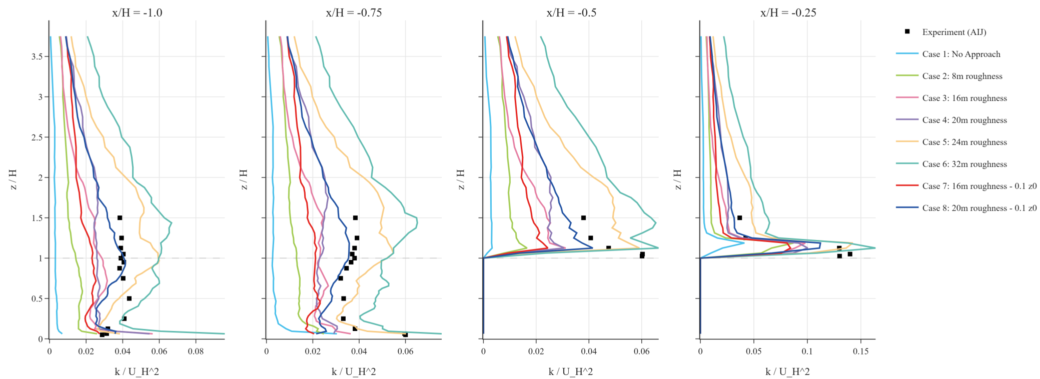


(c)

Figure 13: Normalized vertical velocity fluctuations (w_{rms}/U_H) at (a) Upwind/Rooftop, (b) Leeward Side ($x/H = 0$ to 0.75), and (c) Downwind Wake ($x/H = 1.0$ to 2.0).



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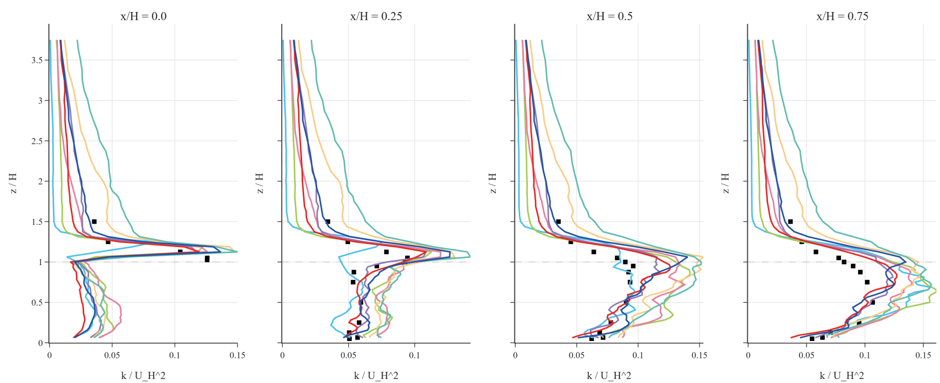

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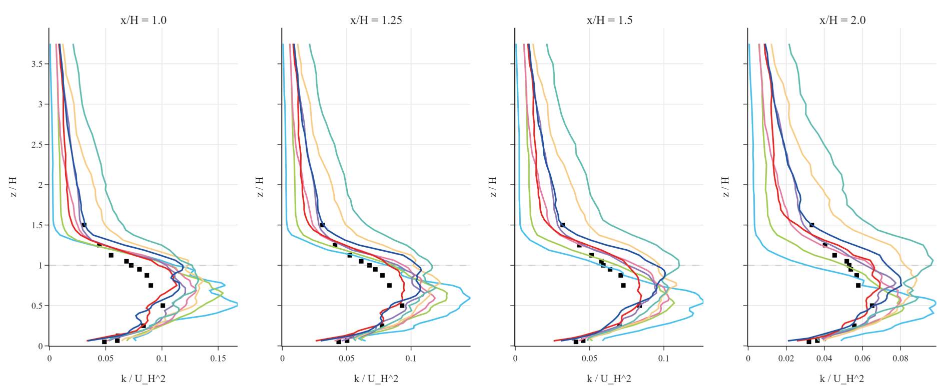


(c)

Figure 14: Normalized Turbulent Kinetic Energy (k/U_H^2) profiles at (a) Upwind locations ($x/H < 0$), (b) Near Wake locations ($x/H \geq 0$), and (c) Far Wake locations.

The profiles exhibit consistent patterns of underprediction in the low-roughness cases and overprediction in the high-roughness cases (Cases 5 and 6). While the absolute magnitudes are smaller than the individual RMS components, the structural characteristics—including the anomalous peaks observed in the no-approach case remain present. The strong similarity between the TKE, v_{rms}, and w_{rms} profiles reinforces the conclusion that the turbulence in the wake is dominated by the lateral and vertical mixing energy. Once again, the medium-roughness configurations, specifically Cases 7 and 8, provide the most balanced representation of the TKE profiles, capturing the energy distribution with the highest degree of fidelity relative to the experimental benchmark.

4.1.2 Validation of Wind and Turbulence Statistics

To quantify the predictive accuracy of the simulation profiles, the LBM results are evaluated using correlation scatter plots and some validation metrics. As detailed in the previous section, the primary indicators used for this assessment are the hit rate (q) and the factor of two observations (FAC2). Through these metrics, a more systematic comparison between the simulated wind speed, fluctuations, as well as the turbulent kinetic energy (TKE), against the reference wind tunnel data can be performed to evaluate the accuracy of the simulation.

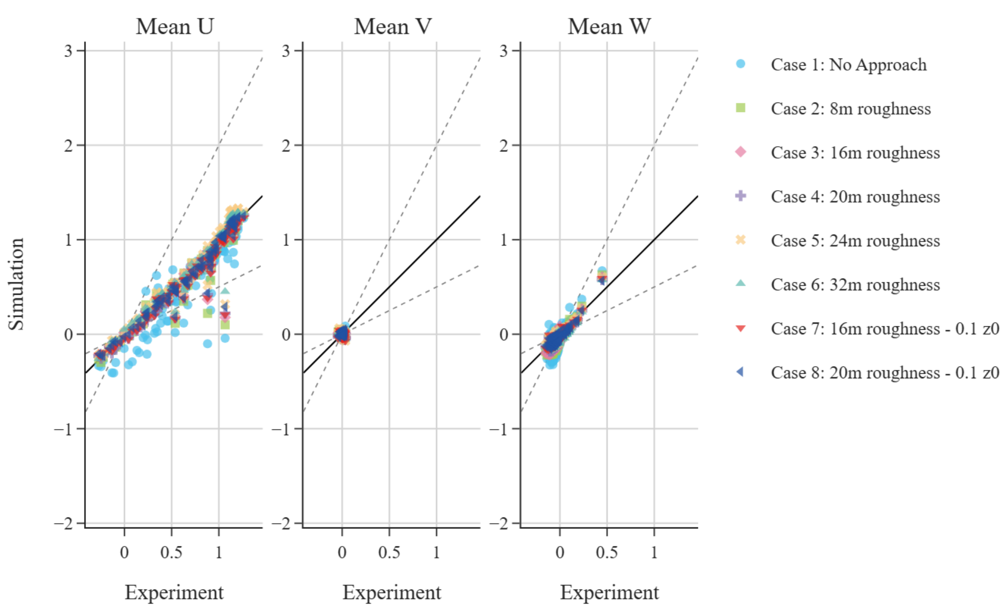


Figure 15: Scatter plot comparison of normalized mean velocity components $(\bar{u}, \bar{v}, \bar{w})$ against AIJ experimental data across all cases. Solid lines represent the 1:1 correlation, while dashed lines indicate the $FAC2$ boundaries.

The correlation scatter plots for the mean velocity components (u, v, w) are presented in Figure 15. Overall, the streamwise velocity (u) demonstrates the highest degree of agreement with the experimental data. In contrast, the spanwise velocity (v) results are difficult to evaluate due to their near-zero magnitudes, and the vertical velocity (w) exhibits a lower correlation compared to the streamwise component.

However, a clear trend is observed regarding the domain configuration, which agrees with the previously seen profiles. The correlation significantly improves in cases featuring medium-height roughness blocks, particularly when the increased surface roughness (z_0) is applied. This suggests that a more developed upwind boundary layer is essential for accurately capturing the mean flow characteristics around the building model, although the inflow turbulence must not exceed a certain limit.

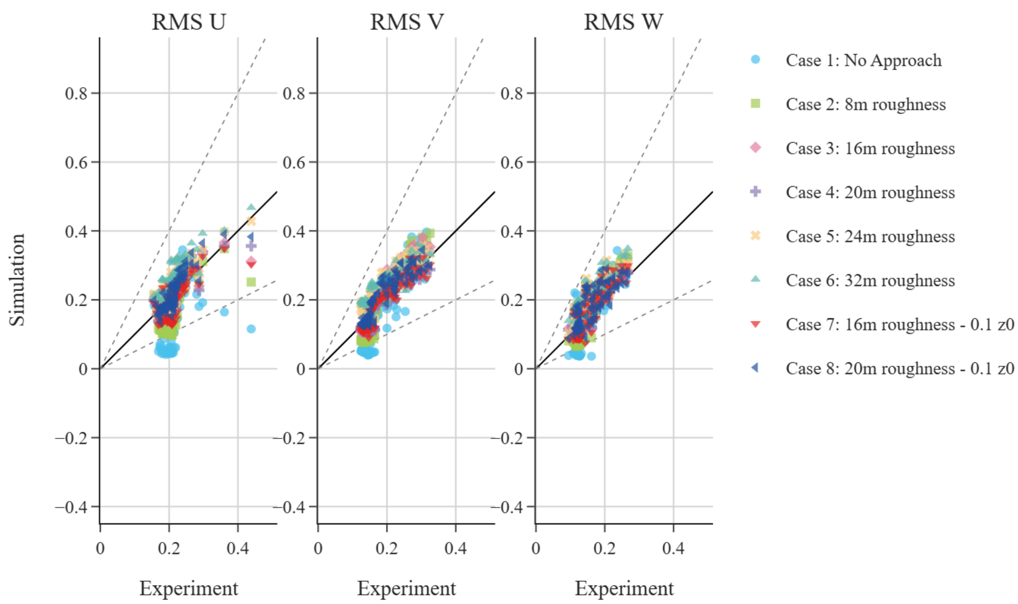


Figure 16: Validation scatter plot for RMS velocity fluctuations ($u_{rms}, v_{rms}, w_{rms}$), demonstrating the solver's ability to capture turbulence intensity across the domain.

The correlation patterns observed in the profile comparisons are further supported by the RMS velocity scatter plots in Figure 16. Unlike the mean velocities, the root-mean-square fluctuations for the u, v, and w components exhibit a high degree of similarity in their distribution. This consistency confirms that the turbulent intensity is captured uniformly across all three dimensions. Furthermore, the trend of increased accuracy in cases with medium roughness blocks is maintained.

The TKE scatter plots (Figure 17) show similar patterns to the trends observed in the RMS velocities, with a consistent increase in accuracy for the medium-roughness configurations. This alignment confirms that the overall turbulent energy distribution in the simulation is fundamentally tied to the development of the approach flow. Specifically, the medium-height roughness blocks appear to provide the optimal turbulence intensity required to match the AIJ experimental benchmarks.

The validation results, summarized in Table 2, more clearly shows the importance of approach flow modeling for achieving experimental agreement. Case 1, which utilizes a local domain without an approach section, consistently fails to meet the $q \geq 0.66$ hit-rate threshold for most parameters. However, the introduction of roughness arrays in Cases 2–8 significantly enhances the development of the turbulent boundary layer, bringing the streamwise velocity (u) and the three fluctuation components (u_{rms}, v_{rms}, w_{rms}) into

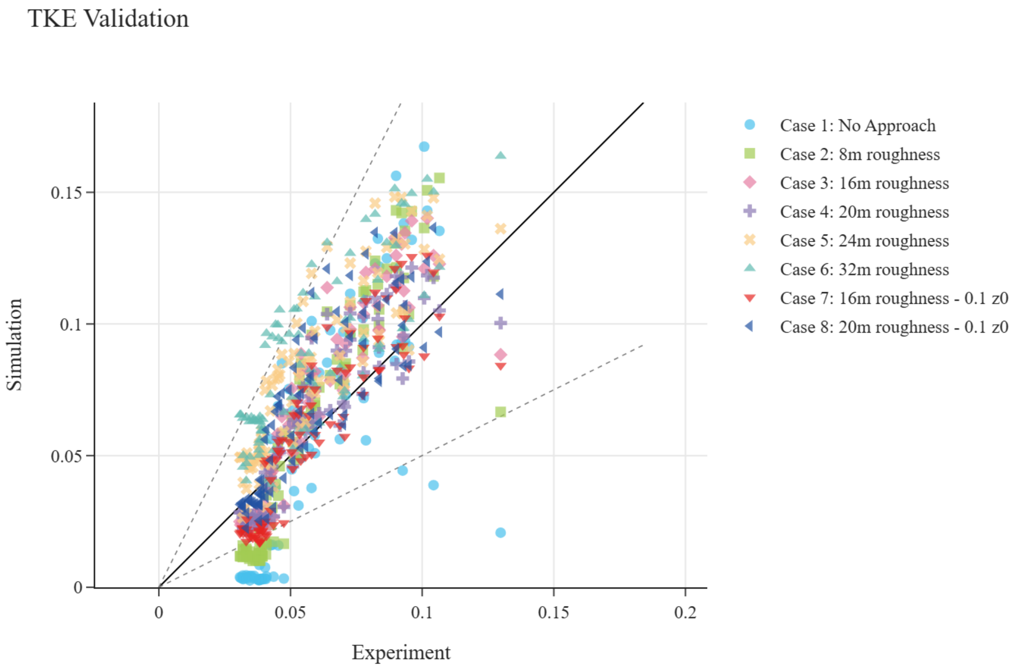


Figure 17: Correlation of predicted vs. observed Turbulent Kinetic Energy (k), highlighting the impact of the refined z_0 and roughness configurations on energy modeling accuracy.

Table 2: Full validation results (q and $FAC2$) for velocity components and turbulence metrics across all approach flow cases. Underlined values indicate performance below the standard thresholds ($q < 0.66$, $FAC2 < 0.50$).

Case	u		v		w		u_{rms}		v_{rms}		w_{rms}		TKE	
	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$
Case 1: No Approach	0.533	0.700	0.933	0.978	0.389	0.433	0.622	0.622	0.644	0.644	0.611	0.667	0.489	0.589
Case 2: 8m roughness	0.900	0.956	0.933	0.989	0.456	0.589	0.944	0.944	1.000	1.000	0.956	1.000	0.611	0.678
Case 3: 16m roughness	0.911	0.967	0.978	1.000	0.533	0.567	1.000	1.000	1.000	1.000	0.956	1.000	0.933	1.000
Case 4: 20m roughness	0.822	0.933	0.900	0.967	0.567	0.600	1.000	1.000	1.000	1.000	1.000	1.000	0.978	1.000
Case 5: 24m roughness	0.900	0.978	0.922	0.967	0.711	0.756	1.000	1.000	0.967	1.000	0.900	1.000	0.656	0.978
Case 6: 32m roughness	0.933	0.978	0.822	0.978	0.578	0.633	0.978	1.000	0.867	1.000	0.856	1.000	0.478	0.833
Case 7: 16m roughness - 0.1 z0	0.867	0.922	0.878	0.978	0.689	0.678	1.000	1.000	1.000	1.000	0.967	1.000	0.922	0.933
Case 8: 20m roughness - 0.1 z0	0.911	0.956	0.967	1.000	0.756	0.644	1.000	1.000	1.000	1.000	0.922	1.000	0.811	1.000

strong alignment with experimental standards.

Similar to the observations in the profile comparisons, the vertical velocity (w) and Turbulent Kinetic Energy (TKE) are the parameters most sensitive to the upwind configuration. While these values occasionally fall below validation thresholds in lower-roughness scenarios (e.g., Case 2), they show significant improvement as the roughness height increases toward the medium-scale configurations. This confirms that the LBM solver, ran on the TSUBAME 4.0 supercomputer, is highly capable of capturing the turbulent structures near a building, provided the inflow turbulence and surface roughness are appropriately set up to match the real conditions.

Table 3: MAE and RMSE validation metrics for velocity components and TKE across all approach flow cases.

Case	u		v		w		u_{rms}		v_{rms}		w_{rms}		TKE	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Case 1: No Approach	0.2009	0.2726	0.0202	0.0267	0.0979	0.1212	0.0759	0.0996	0.0585	0.0675	0.0539	0.0619	0.0268	0.0320
Case 2: 8m roughness	0.0751	0.1518	0.0171	0.0232	0.0585	0.0705	0.0438	0.0575	0.0433	0.0485	0.0394	0.0450	0.0220	0.0254
Case 3: 16m roughness	0.0600	0.1286	0.0133	0.0184	0.0548	0.0650	0.0243	0.0304	0.0354	0.0407	0.0289	0.0341	0.0172	0.0208
Case 4: 20m roughness	0.0690	0.1266	0.0271	0.0321	0.0461	0.0538	0.0232	0.0296	0.0218	0.0259	0.0261	0.0307	0.0123	0.0150
Case 5: 24m roughness	0.0698	0.1194	0.0248	0.0304	0.0354	0.0429	0.0289	0.0366	0.0466	0.0518	0.0408	0.0471	0.0236	0.0286
Case 6: 32m roughness	0.0527	0.0925	0.0290	0.0338	0.0419	0.0501	0.0447	0.0537	0.0470	0.0537	0.0463	0.0524	0.0294	0.0348
Case 7: 16m roughness - 0.1 z0	0.0843	0.1398	0.0247	0.0317	0.0387	0.0459	0.0304	0.0401	0.0215	0.0251	0.0310	0.0366	0.0135	0.0161
Case 8: 20m roughness - 0.1 z0	0.0714	0.1238	0.0145	0.0184	0.0364	0.0415	0.0241	0.0291	0.0256	0.0328	0.0269	0.0377	0.0149	0.0206

The quantitative error metrics shown in Table 3 reinforce the conclusions from the hit rate analysis. In Case 1, the RMSE for streamwise velocity (u) is notably high at 0.2726, indicating a significant local deviation. Conversely, in the roughness-modified Cases 2–8, the RMSE for u is reduced by over 50% in most instances, reaching a minimum of 0.0925 in Case 6.

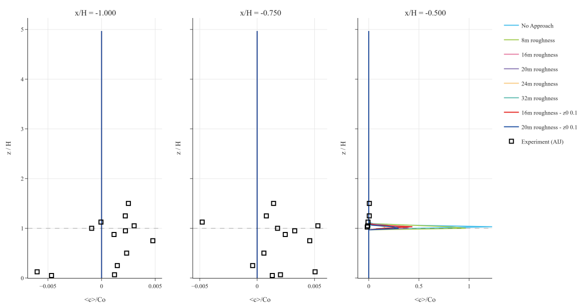
The vertical and spanwise velocity components (v, w) along with turbulence fluctuations maintain stable and low MAE values (< 0.05) throughout most cases. The TKE errors are particularly minimized in the optimized roughness scenarios. This can be seen in Case 4 with an RMSE of 0.0150, demonstrating that again, the solver's ability to capture energy distribution depends heavily on the correct specification of the approach flow. These low absolute errors confirm that the model predictions are both statistically correlated and physically accurate in magnitude.

4.2 Particle Concentration

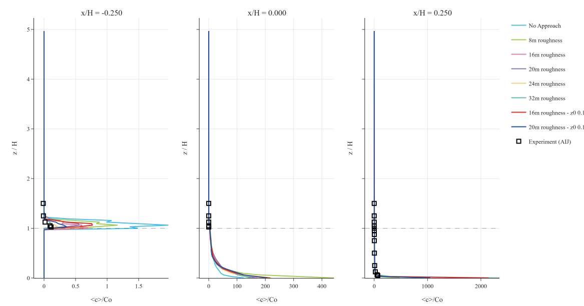
4.2.1 Particle Concentration Profile

4.2.2 Validation of Particle Concentration

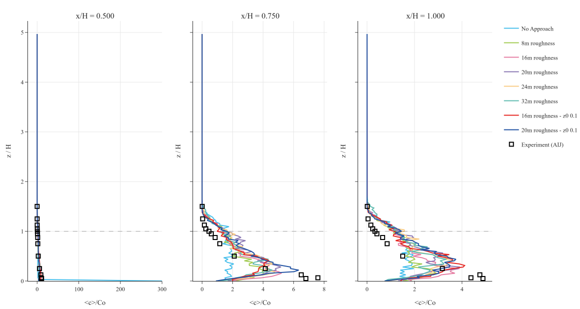
The concentration validation results show a marked contrast to the velocity validation, with significantly lower accuracy across all metrics. For all simulation cases, the hit rate (q) remains well below the 0.66 standard, peaking at only 0.420 in Case 7. While the $FAC2$ values mostly meet the 0.50 threshold (with the exception of the "No Approach" baseline), they are considerably lower than the scores achieved for the flow field. These results suggest that the current Lagrangian particle tracker (LPT) within the LBM solver may not yet fully capture the complex dispersion characteristics of pollutants in an urban environment. This discrepancy could be attributed to the simplified physical modeling of generic particles in the solver, whereas the experimental data were collected using gases that exhibit different diffusion behaviors. Furthermore, the high RMSE observed in Case 6 (9.4304) suggests that even with optimized inflow roughness, the modeling of pollutant concentration remains highly sensitive and requires more specialized treatment of gas-phase



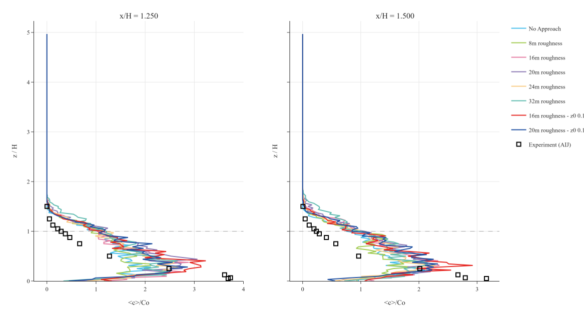
(a) Part 1



(b) Part 2



(c) Part 3



(d) Part 4

Figure 18: Normalized concentration (C_{norm}) profiles across 12 AIJ measurement locations, divided into (a) Part 1, (b) Part 2, (c) Part 3, and (d) Part 4.

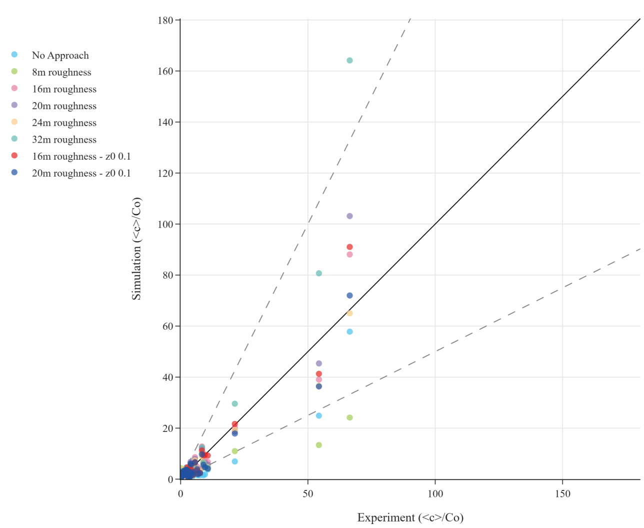


Figure 19: Scatter plot comparison of normalized concentration (C_{norm}) results against AIJ experimental data. The solid line indicates the 1:1 ideal correlation, while the dashed lines represent the $FAC2$ boundaries.

Table 4: Concentration validation metrics across all approach flow cases. Underlined values indicate performance below the standard thresholds ($q < 0.66$, $FAC2 < 0.50$).

Case	**N**	q	**FAC2**	**MAE**	**RMSE**
No Approach	119	0.387	0.496	1.4737	3.6180
8m roughness	119	0.403	0.513	1.6328	5.6510
16m roughness	119	0.395	0.588	1.0878	2.7165
20m roughness	119	0.353	0.504	1.3134	3.7468
24m roughness	119	0.387	0.538	1.0368	2.1463
32m roughness	119	0.353	0.529	2.0231	9.4304
16m roughness (z0 0.1)	119	0.420	0.605	1.0131	2.7927
20m roughness (z0 0.1)	119	0.395	0.538	1.1069	2.2813

physics or turbulent mass diffusion to align with physical observations.

5 Conclusion

Overall, ...

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